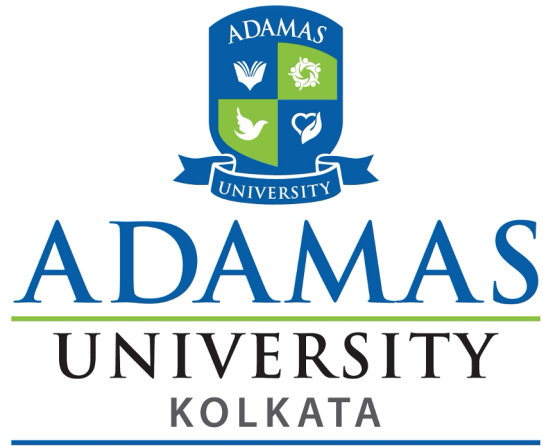




Development of a Machine Learning Model to predict VOC , NOx and Ozone of Urban area of North 24 Parganas

*A Project Report Submitted for Inter Disciplinary Project
Of*

**BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND
ENGINEERING**

Under the guidance of our Project Co-ordinator

Dr. Argha Kamal Guha

CERTIFICATE

This is to certify that the project report entitled “Development of a Machine Learning Model to predict VOC , NO_x and Ozone of Urban area of North 24 Parganas”, is a bonafide record of the work carried out by the team undersigned as part of the *Inter Disciplinary Project*. This report is original and has not been submitted elsewhere for any award or academic credit. The work has been completed under my supervision and guidance. This project report is hereby approved as an original work conducted satisfactorily to warrant its acceptance as a prerequisite to the degree for which it has been submitted. It is implied that by this approval the undersigned do not necessarily endorse or approve any statement made , opinion expressed or conclusion drawn therein, but approve the project report only for the purpose for which it is submitted.

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DECLARATION

This project report is developed and submitted to Interdisciplinary Project of the Bachelor of Technology in Computer Science and Engineering and Civil Engineering under the title of '**Development of a Machine Learning Model to predict VOC, NOx and Ozone of Urban area of North 24 Parganas**' under the title of the project.

It is a genuine report of our personal work that was conducted between the months of September 2025 and November 2025 under the guidance of Dr. Argha Kamal Guha, Assistant Professor, Department of Civil Engineering, Adamas University, Kolkata.

I also testify that I or any of the group members have not presented this project report to receive any award of any degree elsewhere.

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ABSTRACT

Air quality in Kolkata and its neighbouring urban hubs, such as Barrackpore, is gradually worsening over the past years because of the increasing urban traffic density, the concentration of industrial activities and urbanisation. Volatile Organic Compounds (VOCs) are one of the main pollutants compared with other pollutants present in ambient air since they are poisonous, contribute to photochemical smog, as well as, being significant precursors of ground-level ozone formation (Kansal, 2009; Gentner et al., 2017). In that regard, this paper will be devoted to measuring the VOC pollution of the ambient air of the urban site of choice in Barrackpore, and, in this context, the research will attempt to determine the temporal dynamics, potential source impact and reliance of VOC concentrations to the local meteorology.

In this work, we have taken 15-minute resolution data between November 2024 and November 2025 which are taken through the Continuous Ambient Air Quality Monitoring Station (CAAQMS) under the Continuous Airlines Pollution Control Board (CPCB). The VOCs that were taken into consideration were the typical BTEX category Benzene, Toluene, Ethylbenzene, m,p-Xylene and o-Xylene. In addition to them, meteorological parameters of wind speed (WS), wind direction (WD), air temperature (AT), barometric pressure (BP) and relative humidity (RH) were examined. Preprocessing of the data included elimination of the outliers, the treatment of missing data and aggregation of data covering the 15-minute data into hourly and seasonal scales where necessary. Based on this processed data, we studied differences in the diurnal and seasonal change, the VOC-meteorology and we generated wind-rose and pollution-rose plots to determine the prevailing time and direction of contribution to the VOC levels.

The predictive capacity of meteorological inputs to the estimation of VOC concentrations was investigated by training three machine-learning models, i.e., Multiple Linear Regression, Random Forest Regressor and XGBoost. The ensemble models specifically could simulate observed VOC concentrations with quite high precision, which demonstrates that a large portion of VOC variability can be modeled by using routinely measured atmospheric parameters.

According to the outcomes, winter and post-monsoon seasons always have high concentrations of VOCs, primarily, because of worse mixing and more stable atmospheric conditions, and afternoon hours are generally lower because of greater dispersion. Directional analysis indicates some of the traffic-intensive routes and institutionalised emission areas to contribute disproportionately to peak VOC events. In general, the research gives valuable information about the behaviour of VOC in the urban environment of the Barrackpore-Kolkata area as well as the necessity to monitor it constantly, improve the control over the sources of emissions and implement predictive modelling in the system of urban air-quality management.

Keywords: *VOC, Air Quality in the Air, Wind Rose, Seasonal Change, Machine Learning, Urban Pollution, Kolkata.*

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our respected faculty **Dr. Argha Kamal Guha**, Department of Civil Engineering, Adamas University, for his valuable guidance, motivation, and continuous support throughout the completion of this interdisciplinary project titled **“Development of a Machine Learning Model to predict VOC , NO_x and Ozone of Urban area of North 24 Parganas.”** We are also thankful to the **Department of CSE and Department of CE**, Adamas University, for providing the necessary academic environment and resources that helped me complete this work successfully.

We are also expressing our gratitude to Central Pollution Control Board(CPCB) for using their real time air pollutant and meteorological data for this research.

Our heartfelt thanks to my classmates and friends for their cooperation, suggestions, and encouragement during the preparation of this report.

Inter Disciplinary Project Team

B.Tech CSE, 3rd Semester

Adamas University

Date: 24/11/2025

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LIST OF ABBREVIATIONS

Abbreviations	Full Form
AT	Atmospheric Temperature
BP	Barometric Pressure
BTEX	Benzene, Toluene, Ethylbenzene, Xylene
CPCB	Central Pollution Control Board
Ind-VOCs	Industrial VOCs
Comb-VOCs	Combustion-derived VOCs
M-XY	Meta-Xylene
O-XY	Ortho-Xylene
PPB	Parts Per Billion
PPM	Parts Per Million
PPTV	Parts Per Trillion by Volume
RH	Relative Humidity
SR	Solar Radiation
TOL	Toluene
TVOC	Total Volatile Organic Compounds
WS	Wind Speed
WD	Wind Direction
VOC	Volatile Organic Compound

INTRODUCTION

1.1 General

Air pollution has continued to emerge as one of the most chronic environmental issues in contemporary cities. Within the past decades, the quick industrialization, transport network expansion and urban sprawl have all led to the growing emissions of atmospheric pollutants (Heidari et al., 2024; Pandey and Yadav, 2018). Of these contaminants, Volatile Organic Compounds (VOCs) are the most important since they are very reactive, in most cases toxic, and emitted by a broad range of sources. Although VOCs cannot be observed or smelled at the concentrations that they can exist, they may significantly impact human health and the atmospheric chemistry.

VOCs represent a wide family of carbon-based chemicals, which are easily evaporated in the normal environment. They are volatile enough, and they can diffuse through the atmosphere and participate in most photochemical reactions (ATSDR, 2005). Among them, benzene, toluene, ethylbenzene and xylene (or, collectively, BTEX) are of particular interest, as some of them are considered or likely to be carcinogens. The exposure to such compounds over a long-term has been associated with the irritation of the respiratory system, the nervous system, the damage of certain organs, including the liver and kidneys, and the high likelihood of cancer (USEPA, 2020; WHO, 2021).

Both human and natural sources emit the VOCs in and around cities. Vehicle exhaust, dealing with fuel, storing it, industrial, use of solvents, paints, cleaning products including cleaning facilities and commercial processes like filling stations and printing stations are among the major anthropogenic contributors. Natural emissions would primarily be found in vegetation, microbes and plant-soil interactions on the soil. Traffic and mixed urban life are usually the major determinants of the VOC levels in a busy semi-urban area such as the one in the Barrackpore region around the SVSPA campus (Zhu et al., 2013; Abdullah et al., 2014).

VOCs are also vital in the atmospheric chemistry. They contribute to the creation of ground-level ozone (O₃) and secondary organic aerosols which are significant parts of urban smog through sunlight-driven reactions (Hester et al., 1995; Change, 2006). When the wind is slow, there might be temperature inversion and the lack of vertical mixing, these pollutants may accumulate close to the surface and they subject people to exposure.

With such concerns, it is important to learn how the level of VOC varies over time and how it reacts to weather conditions in the immediate surroundings. This paper will discuss the VOC content of the ambient air in one of the chosen urban areas in Barrackpore and discuss the influence of the meteorological conditions on the dispersion of the VOCs; wind speed and direction, temperature, pressure and relative humidity. To explain VOC behaviour and also to determine probable influences of sources, we apply statistical analysis, pollution-rose and wind-rose plots, plus machine-learning-based prediction techniques. Examining seasonal and hourly differences, the work should offer valuable information to the policymakers, environmental agencies and urban planners engaged in the air-quality control and health-protection of the population in this area.

1.2 Ambient Air Quality and Its Impact

Ambient air quality is just a state of air in the outside which we breathe on daily basis. In most cities it is now highly manipulated by the human forces like industrial processes, vehicle traffic, domestic fuel consumption, and the high rate of urbanization though it is affected by the natural processes like wind, sunlight, and rainfall (Pandey & Yadav, 2018; WHO, 2021). The need to sustain clean ambient air has become a significant issue in India and other countries as the population of the urban areas continues to increase and as the energy demand increases.

When there is a build-up of harmful pollutants in the atmosphere a poor air quality is experienced. They are particulate matter (PM_{2.5} and PM₁₀), nitrogen oxides (NO_x) and sulphur dioxide (SO₂), carbon monoxide (CO), ozone (O₃) and some Volatile Organic Compounds (VOCs) (Seinfeld and Pandis, 2016). Some of these pollutants may take long before they settle and others combine with sunlight and other gases to produce secondary pollutants. The traffic of Kolkata and the surrounding areas and Barrackpore, as well as the continuity of construction activity and the proximity to industrial processes and some weather conditions tend to create the periods of high pollution levels.

Poor quality air in the environment has environmental and health consequences. In the environment, high levels of pollutants may cause smog, low visibility, acid rains and destruction of vegetation and crops. The significance of VOCs lies in the fact that they play an active role during the process of formation of ground-level ozone and secondary organic aerosols, which exacerbate urban smog and may disrupt the functioning of plants, including photosynthesis and respiration (Hester and Harrison, 1995). On health grounds, long run exposure to polluted air causes the risk of respiratory diseases such as asthma and bronchitis, cardiovascular and nervous system and in other instances, predisposes the chance of cancer. Typically, the populations most vulnerable include children, older adults and people with underlying medical issues (USEPA, 2020; WHO, 2021). The short term effects of the high levels of VOCs and other gases include headaches, throat and eye irritation, fatigue and limited lung capacity.

The weather conditions have a significant effect in the behavior of pollutants. Such parameters as temperature, relative humidity, wind speed, wind direction, and wind pressure define whether the pollutants will be dispersed or not or will be trapped at the surface. As an illustration, when the winds are slow or the temperature inversion situation takes place in winter, the pollutants will be at a lower altitude and may result in a sudden increase in concentration, which is common in post-monsoon and winter seasons in eastern India (Gong and Wang, 2017).

Due to these issues, one should learn more about the ambient air quality and learn the way in which pollutants depend on the meteorological factors. It is in this project that VOC measurements are combined with meteorological data to get a better understanding of the air-quality situation in a semi-urban area such as Barrackpore. The method can also assist in determining critical periods and high exposure conditions and assist in planning of air-pollution management and environmental health management in the area better.

1.3 VOC–Meteorology Nexus

Volatile Organic Compounds (VOCs) do not act as individual behavior in the atmosphere. Weather conditions have an overwhelming effect on their concentration and movement. This complex of VOCs and atmospheric processes has been commonly described as the VOC-meteorology nexus. Simply put, the level of VOCs to be found at a given location at a given moment in time is a result of the behavior of the atmosphere such as the air temperature, speed of the wind, moisture content and the stability or instability of the air column (Atkinson, 2000; Seinfeld & Pandis, 2016).

The most important of the various meteorological conditions are the wind speed (WS) and the wind direction (WD). Under the circumstances of extremely low wind speed, air is almost stagnant and VOCs are likely to be concentrated in the lower part of the air. With a higher wind speed, the number of pollutants carried throughout a larger area and then the concentration of the pollutants decreases. Wind direction gives information on the origin of the air mass, hence the VOCs. Wind-rose and pollution-rose maps can commonly have evident directional arrangements that can be linked with traffic routes, industrial and mixed-city sources (Dimitriou and Kassomenos, 2015).

The same but not similar effect is air temperature (AS). Increased temperature is favoring evaporation of fuels and paints and other organic substances, hence the growth in VOC emissions. Photochemical reactions are also accelerated when it is warmer (particularly during sunny days), and this may convert VOCs as well as increase ozone formation. These reactions are caused by solar radiation (SR), which is used to convert VOCs and NO_x into ozone and secondary organic aerosol (Gentner et al., 2017; Guo et al., 2020).

Barometric pressure (BP) and relative humidity (RH) have more of an indirect effect. Some photochemical pathways may be suppressed by very humid conditions and more efficient ozone production is commonly favoured by drier conditions. The atmospheric stability and vertical mixing are influenced by changes in pressure. The high-pressure systems tend to create more stable conditions and thin layers of mixing that can entrap the pollutants near the surface and cause high concentrations (Seinfeld and Pandis, 2016).

These forces can be more evident upon examining seasonal and daily trends. When cold air in the eastern side of India and the Barrackpore area, in winter, warmer air mixes with the lighter air and the low mixing heights tend to trap the pollutants and VOCs on the surface. Conversely, the monsoon and summer winds are stronger and the rainfall and increased convective mixing dilute the pollutants more (Gong and Wang, 2017; Wang et al., 2019). Even during a day, the VOCs tend to increase in the morning during the traffic peaks in the morning and decrease in the afternoon, when the atmosphere is more unstable and vertical mixing is enhanced.

This VOC-meteorology association is of importance to study due to the fact that it can be used to understand how and why the level of pollution can be higher and it can be used to predict and model the concentration of VOC. By examining the influence of temperature, humidity, wind speed, wind direction and pressure on VOC levels at Barrackpore in this project, we can be able to understand the given observations of pollution patterns in a better and likely be able to justify a data-based air-quality management of the local urban environment.

1.4 Machine Learning(ML) Application for Air Quality Model

Machine learning (ML) is fundamentally the tool of choice when exploring air-quality information, particularly when the pollution energy is of any nature nonlinear, as well as affected by such factors as temperature, wind velocity, humidity, pressure, and human activity. The simple type of the model of traditional stats are cool in the straightforward things but complete failures in cases where the levels of the pollutants are conditional on a mash-up of the factors. In fact, ML models may acquire such complicated patterns in the past - that is what the papers by Chen and Zhang (2020) and Xiong et al. (2021) are all about.

With regards to VOCs and other gaseous pests, ML will allow us to deconstruct the ways all the meteorological parameters push the levels of these substances either uphill or downhill. To take some examples, a model could calculate that during a quiet night in winter, a slight alteration in temperature implies an increase in VOC accumulation, or that intense winds destroy pollution in the afternoon. It is also able to record the rapid, hourly oscillations, which is important to assemble mornings and evenings peaks noticed by all in urban and suburban areas (Dimitrova and Hirabayashi, 2021).

The standard ML tricks to predict air-quality are Multiple Linear Regression, Decision Trees, Random Forest, Gradient-boosted beasts such as XGBoost, and all the various Neural Networks peculiarities. Typically we tend to train them by utilizing previous pollutant measurements and meteorological measurements such as air temperature (AT), relative humidity (RH), wind speed (WS), wind direction (WD) and barometric pressure (BP). Once trained, new conditions can be imposed, and estimated levels of pollutants are obtained - extremely helpful to researchers and policymakers to identify high-pollution periods (Peng and Sun, 2022).

One of the selling-points of most ML algorithms, and of the tree based ones, in particular Random Forest and XGBoost, is that they provide you with feature importance. That is what you know which of the inputs causes the biggest influence on the output. In the case of the VOC, you may discover that the large players are temperature and wind and humidity and pressure are the side-kicks. The latter are gems of eco-planning and creation of early-warning mechanisms that target the most delicate variables.

It will involve machine learning in this project to research VOC behaviour at the Barrackpore CAAQMS location and construct predictive models based on the most influential meteorological variables. Through the training and validation of models like Multiple Linear Regression, Random Forest and XGBoost, we hope to have a better knowledge of the interactions of the pollutants with the weather, and to develop a realistic model of predicting the level of VOCs in the Barrackpore region.

1.5 Research Aim

The primary objective of the project is to calculate the way VOCs change in the outdoor air surrounding Barrackpore and the way their concentrations vary depending on weather changes. VOCs are not cleanly distributed, but rather poorly distributed by temperature, humidity, wind speed, wind direction, and pressure; and, therefore, we are excavating how each of them pulls concentrations of the VOCs.

The second goal is to develop and experiment with machine-learning models that will predict the VOC levels based on the accessible weather data. Training the models on previous VOC and weather data of the CAAQMS station, Barrackpore, will answer whether the ML method can provide us with any sensible predictions.

In general, the study will integrate air-quality monitoring and meteorological assessments with ML-powered prediction to create a more accurate image of the dynamics of VOCs in such a semi-urban area as Barrackpore. What we would like to do is to put together a fundamental scientific understanding which can be used to plan on how to improve and better the level of control of emission-control.

The final objective is to produce a supporting science, which will be able to facilitate better planning, control of emissions, and future air-quality management of the region.

1.6 Novelty Statement

The originality of this work is that the high-resolution VOC measurements were combined with local meteorological data of the Barrackpore area, and the relationship between the two was subsequently investigated with the help of the Machine Learning in a very simple and practical manner. Although much of the air-pollution research is done on particulate matter, or broad-based measures of air-quality, not much is done on how VOCs respond at a semi-urban scale such as at Barrackpore or how they react to daily weather conditions in this area.

The other distinctive feature of the work is that it has a small set of meteorological parameters that are carefully chosen in order to create prediction models: air temperature, relative humidity, wind speed, wind direction and barometric pressure. The study demonstrates that despite the need to use large and complex input datasets to estimate the levels of VOC, even a small set of regularly available variables can be utilized to estimate the levels of VOC with a reasonable level of accuracy using models like the Random Forest and XGBoost.

Another novelty of the project is the collaboration of the investigation of seasonal and hourly differences in VOCs and their correlation with wind-rose analysis and time-slot analysis. This gives a better understanding of how the VOC pollution action is during various times of the day and various seasons. In general, the methodology is a combination of persistent surveillance, simple statistical and machine-learning predictive models to develop an effective framework that can assist in the future research and air-quality management efforts in and around Kolkata.

1.7 Organisation of the Report

This project report is divided into six chapters so that the study can be followed in a clear and logical sequence, starting from the background and ending with the final findings and future scope.

Chapter 1: Introduction

This chapter presents the background of the study and explains why ambient air quality and VOC analysis are important. It includes a brief discussion on ambient air pollution, the VOC–meteorology nexus, the role of machine learning in air-quality modelling, the research aim and objectives, the novelty and scope of the work, and an outline of the overall structure of the report.

Chapter 2: Literature Review

This chapter summarises previous studies related to VOC pollution, their major sources, health and environmental impacts, and the influence of meteorological factors. It also identifies the research gaps in existing work and explains how the present study attempts to address some of those gaps.

Chapter 3: Research Methodology

This chapter describes the methods adopted to carry out the study. It introduces the study area and dataset, lists the pollutant and meteorological parameters used, and explains the steps of data preprocessing and analysis. The tools, software and statistical techniques are outlined, along with the Machine Learning models developed for VOC prediction.

Chapter 4: Results and Discussion

This chapter presents the main results of the analysis, including VOC concentration levels, diurnal and seasonal patterns, time-slot and directional behaviour, and their relationship with meteorological parameters. It also discusses the performance of the Machine Learning models and interprets the findings in the context of the study area.

Chapter 5: Findings and Conclusion

This chapter summarises the key findings of the research and discusses their environmental and practical significance. It presents the main conclusions drawn from the work, outlines the limitations of the study and suggests directions for future research and possible applications.

Chapter 6: References

This chapter lists all the books, research articles, reports and online sources referred to during the preparation of the project report, formatted according to the chosen referencing style.

LITERATURE REVIEW

2.1 General

Volatile Organic Compounds, also referred to as VOCs, are a significant issue in air pollution that we are undertaking as environmental scientists, and they have a serious impact on the quality of the air that surrounds us. They come out of such daily sources as car emissions, gasoline and diesel emissions, paint, solvents, small-scale industrial facilities, as well as domestic heating (Kansal, 2009; Gentner et al., 2017). The combination of the traffic, local workshops, local shops, and houses can increase the level of VOCs in such locations as the Barrackpore area around the SVSPA campus significantly.

VOCs are not only toxic, but they also participate in other chemical processes in the atmosphere. It is upon sunlight and increased temperatures, VOCs react with nitrogen oxides (NO_x), resulting into photochemical smog as well as the formation of ground-level ozone (Hester and Harrison, 1995; Seinfeld and Pandis, 2016). Some of the VOCs like benzene and other aromatics are known to have health hazards particularly when exposed over a long duration of time. We experience such symptoms as irritation of eyes and throat, respiratory difficulties, dizziness, and severe cases are reported as a heightened risk of cancer (ATSDR, 2005; USEPA, 2020; WHO, 2021).

Volatile Organic Compounds (VOCs) have a close relationship with weather. Such aspects as temperature, relative humidity, and wind speed, and wind direction assist in determining whether VOCs disperse, are transported, or remain close to the surface. Indicatively, when wind speed is low and the layers are stable, VOCs are able to accumulate near the ground, but on high wind speeds and due to unstable conditions, they can be diluted and transported to distant locations (Gong and Wang, 2017). This is the reason why we should pay attention not only to sources of emissions but also to meteorological conditions when researching the pollution of VOC.

Over the past years, VOCs have been given more attention due to their twofold effects on the environment and health. The most important stage in designing improved air-pollution controls is to monitor their concentrations, identify the key sources and observe how they vary with weather changes. It is in this section that the minimum background information needed to comprehend why VOCs are important would be provided and it would form a focal point in the study of urban air-quality in locations such as Barrackpore.

2.2 Urban Air Pollution and Its Health Consequences

Cities and suburbs are characterized by the apparent problems with air quality due to the increased number of vehicles, local enterprises, building and building maintenance, as well as residential heating (Pandey and Yadav, 2018). Although the pollution is not that congested as in central Kolkata, the area surrounding the Swami Vivekananda State Police Academy (SVSPA), Mangal Pandey Uddyan, is still polluted to some extent due to the surrounding traffic routes, the constant bus routes, and weather conditions which occasionally slow down the distribution of the pollutants.

That pollution includes VOCs (volatile organic compounds). They are produced by the car exhaust, gasoline fumes, paint, solvents, and housework (Kansal, 2009; Gentner et al., 2017). Traffic of buses, autos, scooters, and personal cars passing in the surroundings of Academy is a constant source of VOCs. In cases of the cold wind or when the air remains stagnant, such chemicals remain near the ground longer, increasing exposure to trainees and teachers, staff, and the local residents (Gong and Wang, 2017).

Air pollution in urban areas is a nuisance to the environment and health. The presence of sunlight and nitrogen oxides in the environment makes VOCs contribute to the formation of ground-level ozone and photochemical smog, which reduces visibility and may cause irritation of our lungs (Seinfeld and Pandis, 2016). Breathing of such air over a long period of time has been associated with coughing, breathing discomfort, eye and throat irritation, headaches, asthma and decreased lung capacity. Children, adults, and those with breathing issues who already have them are generally at the greatest risk (USEPA, 2020; WHO, 2021).

Some VOCs are particularly ugly such as benzene, toluene and xylene. They may influence central nervous system and with time may increase the chances of cancer (ATSDR, 2005). Besides outdoor sources, there are certain indoor items such as cleaning chemicals, maintenance materials, and cooking fuels that contribute to the total amount of VOCs exposure to individuals located inside the campus buildings.

It is essential to study the VOCs behaviour around SVSPA, Mangal Pandey Uddyan, Barrackpore. It assists in identifying the period when pollution is spiking and increases awareness to all people in campus. Such analysis data may be used to design monitoring plans and enhance environmental-health management of both the Academy and the adjacent neighborhood.

2.2.1 Indian Scenario of VOC Pollution

The release of Volatile Organic Compounds (VOC) into the atmosphere is steadily increasing due to the large number of motorized vehicles, and large-scale growth of manufacturing and building projects in India. There are many types of sources of VOC emissions including petroleum and diesel automobile exhaust, chemical processing, paint and coatings, oil refineries, small workshops and household fireplaces or fires to burn trash (Kansal, 2009; Pandey & Yadav, 2018).

As both urban and semi-urban areas expand, managing VOC levels has become a growing concern to the urban air quality management community. Studies have identified VOC's in urban atmospheres across multiple Indian cities, including Delhi, Mumbai, Kolkata, Chennai, Hyderabad and Bengaluru, specifically benzene, toluene, ethylbenzene and xylenes (BTEX), by Kansal (2009) and Sahu & Saxena (2020). The concentration of BTEX in urban environments, particularly in cities like Delhi and Mumbai, is commonly attributed to heavy vehicular traffic and an abundance of industrial facilities, along with evaporative losses from vehicle fuel storage and delivery systems. High BTEX concentrations were also found throughout the city of Kolkata, through multiple studies, and they were highest during the winter months when the temperature is cold enough to produce poor vertical mixing of the atmosphere. This can cause the pollutants that are released into the atmosphere to remain close to the ground, thus creating high concentrations of VOC's.

Monitoring of VOC by the Central Pollution Control Board (CPCB) and State Pollution Control Boards under the National Air Quality Monitoring Programme (NAMP), has increased continuously over time via selected CAAQMS stations. Nevertheless, there continues to be much less widespread VOC monitoring than for PM and gases like NO_x and SO_x. Therefore, VOC data is available only at a very small number of sites at present and so the assessment of long term trends in VOC across all of India is highly problematic.

VOC levels in most cities in India tend to be elevated in the winter season primarily due to temperature inversions, restricted vertical mixing, and weak wind patterns that impede pollutant dispersion (Mukherjee & Agrawal, 2020). VOC levels during the warmer months and during the monsoon season, generally decrease primarily due to enhanced convective mixing and greater precipitation frequencies, but local VOC emissions can lead to periodic peaks in VOC concentrations. Poorly maintained vehicles, traffic congestion, and poor fuel quality contribute to additional VOC emissions in many locations.

As VOC's negative impacts on human health become increasingly recognized — particularly benzene, identified as a human carcinogen — this topic is becoming increasingly critical. Many studies have documented symptoms including headache, dizziness, upper respiratory tract irritation, and possible long-term health risks due to VOC exposure in high VOC concentration areas (WHO, 2021; ATSDR, 2005; USEPA, 2020). As a result, the CPCB and other regulatory bodies have initiated efforts to identify VOC emitting sources, develop emission inventories, and impose stricter regulations on VOC emissions from industrial processes (CPCB, 2022).

Therefore, based on the above, VOC pollution appears to be a significant growing concern in India. With continued urbanization and industrialization, the investigation of VOC behavior, identification of VOC emitting sources, and analysis of how VOC concentrations respond to weather phenomena will be necessary in order to improve ambient air quality and protect human health.

2.2.2 VOC Scenario in West Bengal

Studies on VOCs in West Bengal have been gaining momentum over the past few years especially in areas with heavy traffic volumes and local industrial activity. High VOC concentrations have been identified in cities and towns throughout West Bengal including Kolkata, Howrah, Durgapur, Haldia, Barrackpore and Asansol from vehicle emissions, fuel handling at petrol stations, small scale industry, paint shops, vehicle repair, burning of solid waste and use of fuels for cooking (Pandey & Yadav, 2018; Kansal, 2009). The types of emission sources identified are common to numerous developing urban and semi-urban centers in West Bengal.

The West Bengal Pollution Control Board conducts air quality monitoring at a number of stations within the State of West Bengal, however there is currently limited ongoing VOC monitoring relative to other monitored parameters such as PM_{2.5}, PM₁₀, SO₂, and NO₂. As a result, most data related to VOC's in West Bengal has been generated through short-term field campaigns, academic research projects and university/ environmental institute-based surveys (WBPCB, 2022) for a number of years. The majority of the studies conducted on VOC's were focused on BTEX (benzene, toluene, ethylbenzene and xylene) since they are commonly used indicator compounds of VOC urban air quality and have a direct relationship to public health.

VOC concentrations tend to be greater in the vicinity of busy street intersections, markets, industrial facilities and gas stations in and surrounding the cities of Kolkata and Howrah. VOC levels are typically greatest in winter due to cooler temperatures, smaller mixing heights, and lesser wind speeds resulting in VOC's being trapped near the ground. Conversely, VOC levels tend to be lower during the monsoon season with increased precipitation and improved vertical ventilation resulting in greater removal of VOC's. However, localized VOC increases can still occur in the vicinity of strong VOC emitting activities (Mukherjee & Agrawal, 2020).

VOC levels in suburban or semi-urban areas such as Barrackpore, Khardah, and Kalyani will vary based upon distance to a primary transportation route, local commercial district(s), and level of residential burning practices. While VOC levels in suburban areas may be lower than those found in central Kolkata, VOC emissions from two-wheeled vehicles, autos, buses, and small workshops remain significant, particularly during periods of heavy vehicle usage.

Concern regarding the presence of benzene in fuel vapor and exhaust emissions is elevated in West Bengal due to benzene's known carcinogenic properties (WHO, 2021; ATSDR, 2005). As a result of this concern, WBPCB and research institutions working collaboratively with WBPCB have emphasized additional research into VOC source characterization, diurnal and seasonal behavior and the impact of meteorological conditions on short-term VOC variability (WBPCB, 2022). The increasing interest in VOC pollution indicates that VOC pollution is not only a problem for large metropolitan areas but is also significant in suburban and semi-urban areas throughout the State of West Bengal.

Therefore, understanding VOC conditions in West Bengal provides a critical regional perspective for the current study and is especially relevant to the SVSPA campus at Mangal Pandey Uddyan, Barrackpore, where local emissions and meteorological conditions together shape the overall quality of outside air.

2.2.3 VOC Scenario in Overseas Countries

In simple terms, VOCs make a significant urban air-quality policing issue since they contribute to smog, ground-level ozone, and chronic health issues (WHO, 2021; Gentner et al., M. 2017). Avoiding the use of solvents, refineries, fuel centers, and vehicles, people in wealthier countries such as the USA, Japan, Australia, Canada, and the EU states have been monitoring VOCs over decades and imposed restrictions on solvents, refineries, fuel centers, and vehicles, all because of the role of VOCs in causing ozone and organic particles in the urban air (USEPA, 2020; EEA, 2022).

According to U.S. statistics, where there are traffic jams and petrochemical areas, VOC numbers tend to be the highest; the ozone protection plans bind VOC reductions directly to the achievement of regional targets (USEPA, 2020; Seinfeld and Pandis, 2016). Tougher fuel regulations, superior vapour collection, LDAR regulations and solvent rules reduced NMVOC emissions across Europe over the past 20 years, yet winter weather and rush hours continue to spur up higher BTEX in giant cities (EEA, 2022; WHO, 2021).

In some areas of Asia that happens to be urbanizing at a very fast rate, folks observe VOCs bonded to the emergence of cars and industries, which are fond of solvents. Large cities in China such as Beijing-Tianjin-Hebei and the Yangtze Delta registered highly reactive VOCs that cause objectionable winter smog, thus, implementing tighter rules of industry and solvents besides increasing stricter fuel losses (Guo et al., 2020; WHO, 2021). In Japan and South Korea, early, harsh restrictions on the sale of paints, solvents, and refineries by the VOC and the air in the city were certainly better (WHO, 2021). Oil-and-gas plants and hot weather, which increases evaporation velocity in Middle-Eastern petrochemical centers, maintain VOC at a high level (Gentner et al., 2017; WHO, 2021). On the other hand, lower urban VOCs are typical of Australia and New Zealand, only soaring in industrial areas and on hot summer days (WHO, 2021).

All this global stuff, such as winter hold -up, rush -hour spikes, and directional trends, can provide us with a yardstick and explain the VOCs in India. It preconditions our current Barrackpore (SVSPA) research, whereby mixed traffic lanes, seasonal consistency, and day-to-day weather conditions are expected to adjust VOC levels in the same way that we observe them in other metros across the globe (Seinfeld and Pandis, 2016; WHO, 2021).

2.3 Sources of VOCs

Human activities are the dominant source of VOCs in cities and the surrounding environment, although in the latter case, natural factors are also significant (Gentner et al., 2017; Pandey and Yadav, 2018; WHO, 2021). Traffic is the largest of all sources - BTEX (benzene, toluene, ethylbenzene, xylenes) is increased by tailpipe emission and the evaporative losses of refuelling or fuel system leakages, which are especially important in two-wheelers, older cars that lack vapor-control technology and other larger vehicles (USEPA, 2020; EEA, 2022; Kansal, 2009).

The intended industrial and craft environment emits a lot of VOCs when it uses solvents - consider paint or coating shops, printing, adhesives and metal finishing, in addition to other chemical manufacturing. When the ventilation is low, and the leakage-detection programmes (LDAR) is not large enough to collect fugitive leaks, then the loads become even higher (Gentner et al., 2017; WHO, 2021; EEA, 2022). Storage and distribution of fuel, such as pumps of petrol, depots or LPG, introduce additional emissions due to inhalation by the tank and loading processes as well as leakage of the lines (USEPA, 2020; CPCB, 2022). Domestically, there are peaks of VOC due to cleaners, repellents, aerosols, fresh paint and varnishes, and the burning of biomass or waste (leaves, firewood, trash) (Pandey and Yadav, 2018; CPCB, 2022; WHO, 2021).

Plants of course do produce isoprene and monoterpenes, and these levels explode on hot sunny days. Even those biogenic VOCs are important in the formation of ozone and secondary aerosols (Seinfeld and Pandis, 2016; Guo et al., 2020).

In the environs of SVSPA, Mangal Pandey Uddyan (Barrackpore), there is that real source mix of constant traffic and refuelling ripples up the figures, some of that domestic solvent use in small workshops, domestic, and the burning at times - not to mention that natural background of vegetation. The awareness of the classes that contribute to the patterns (various wind directions or times of the day when peaks are observed, etc.) will enable us to select the appropriate control options (CPCB, 2022; WHO, 2021; EEA, 2022)..

2.4 Meteorological Influence on VOCs

The overall meteorological effects have a significant effect on the VOCs dynamics once released. They vary on an hourly and a seasonal basis based on factors such as temperature, sunshine, winds speed and direction, relative humidity and stability or pressure of the atmosphere as a whole. These processes regulate the movement, distribution, dilution, and chemical reaction of the pollutants (Seinfeld, and Pandis, 2016; Atkinson, 2000).

Heat and sunlight influence the process of emitting and reacting of VOC. Warmer climate causes fuels, paints and solvents to evaporate faster whereas more actinic light accelerates the photochemistry of the VOC-NO_x. It is more likely to increase the production of ozone and secondary organic aerosol during hot and sunny days (Gentner et al., 2017; Guo et al., 2020).

The speed and direction of the wind are vital in the realization of how pollutants spread and their sources. Slow wind speeds accumulate pollutants around roads and refuelling points, but greater speeds transport the pollutants. Wind rose or pollution rose analysis can be used to trace the peaks of pollution to the upwind sources and areas of traffic (Dimitriou and Kassomenos, 2015; EEA, 2022).

The relative humidity alters the reaction pathway through alteration of radical budgets, as well as, the distribution of gases between the air and the particles. Most photochemical reactions tend to occur efficiently when the air is dry, whereas some photochemical reactions are inhibited under extremely wet conditions and the partitioning of VOCs can change under such conditions (Seinfeld and Pandis, 2016; WHO, 2021).

The vertical mixing is curtailed by pressure and stability, as inversions or shallow boundary layers beneath anticyclones, which retain the pollutants near the surface. On the other hand, when the pressure is low or the air layer is convected, the mixed air layer is extended and pollutant concentrations are weakened (Jacob, 1999; Gong and Wang, 2017).

Such controls are evident in pattern seasonal and daily patterns in eastern India. During conditions of low mixing and mono- or poly- air, the levels of VOC are expected to be high in winter or after the monsoon. Stronger winds, rainfall and enhanced ventilation lower concentrations during the monsoon and summer. Within a day, the peaks of pollution are frequently coincidental with morning and evening traffic when there is no mixing, and afternoon is typically clean because of the convective turbulence (Mukherjee & Agrawal, 2020; Pandey and Yadav, 2018).

In the case of the SVSPA campus in Mangal Pandey Uddyan, Barrackpore, analysis of VOC data in a meteorological perspective (i.e., the speed and direction of the wind in dispersion and source direction, temperature and solar radiation in emission and reactivity, and relative humidity and pressure in atmospheric stability) plays a role in understanding why the levels of pollutants vary by season and by hour. Such a viewpoint allows making more accurate predictions and managing the quality of air locally more efficiently (CPCB, 2022; WHO, 2021).

2.5 Effects of VOCs on Environment and Human Health

Human exposure to high levels of Volatile Organic Compounds (VOCs) has been shown to have a variety of adverse effects on human health. Short term exposure to high VOC levels results in eye, nose and throat irritation, headaches, dizziness, and fatigue as well as exacerbating the symptoms of individuals who have asthma or other respiratory problems (USEPA, 2020; WHO, 2021). In addition to the short term adverse health effects that result from exposure to high levels of VOCs, long term exposure to certain types of aromatic compounds such as benzene, toluene, and xylenes (BTEX) may result in central nervous system effects and decreased lung function. The World Health Organization has classified benzene as an established carcinogen and has identified vulnerable populations such as children, older adults, and those with existing cardio-respiratory diseases as being at increased risk for the adverse health effects resulting from exposure to VOCs (WHO, 2021). VOCs are emitted indoors by common products used in daily life including fresh paint and coatings, cleaning agents, aerosols, and chemicals that are stored and used in the home. Indoor VOC levels may become extremely high in homes without proper ventilation and indoor VOC exposures will be additive to outdoor VOC exposures experienced throughout the day (USEPA, 2020; WHO, 2021).

The impact of VOC emissions to the environment and human health will likely be most pronounced in and around the Swami Vivekananda State Police Academy (SVSPA) located in the vicinity of the Mangal Pandey Uddyan, Barrackpore during periods when there is heavy traffic volume and stagnant meteorological conditions. It is important to recognize the environmental and health related pathways through which VOC emissions occur in order to develop strategies to minimize VOC emissions in the campus neighbourhood (WHO, 2021; USEPA, 2020). Routine VOC monitoring, educating the public about VOC exposure, and implementing measures to prevent VOC emissions will be critical steps to reducing VOC exposures in the area surrounding the SVSPA.

2.6 ML Models for VOC Forecasting

In recent years, many researchers have started using Machine Learning (ML) techniques to forecast VOC concentrations because traditional statistical models often struggle to capture the non-linear relationships between meteorological factors and pollutant behaviour. Studies from India and abroad have shown that ML models such as Random Forest, Gradient Boosting, Support Vector Regression, and Artificial Neural Networks can provide more reliable predictions, especially when multiple weather parameters are involved.

Different studies have used different combinations of VOC species, meteorological inputs, and ML algorithms depending on their research location and data availability. Most of these works report performance using the **R² score**, which helps us to understand how good this model explains variation in pollutant concentration. The following table summarises a few commonly referenced studies that employed ML for VOC or air-quality forecasting.

Table 2.1 — Summary of ML Models Used in Previous VOC Forecasting Studies

Year	Study / Location	ML Model Used	Reported R ²	Notes (what was predicted)
2021	Rezaali et al., Indoor BTEX, Iran (PubMed)	Wavelet-based Random Forest (WRF)	0.98 (test)	Indoor BTEX concentrations using temperature, RH, day-of-week, location as inputs.
2021	Nkeshita et al., Univ. Hostel, Nigeria (ResearchGate)	Cascaded Feedforward ANN	0.98	Indoor Total VOC (TVOC) in hostel room, 6 inputs incl. PM, RH, Temp, HCHO.
2025	Wang et al., Taiwan (Jung group) (ResearchGate)	Random Forest with Land-Use & Met Variables	0.85 (max CV)	Outdoor BTEX at 1 km resolution; cross-validation R ² for some BTEX species (o-xylene, m,p-xylene) $\approx$ 0.85.
2025	Eid et al., Multi-Output VOCs, paint workers (Najah Staff)	Multi-output Gaussian Process (MOGP)	0.94 ($\approx$ 0.9355)	Predicting multiple VOCs (e.g., acetonitrile) from co-emitted VOCs + demographic data.

2.7 Summary of Literature Review

The findings of the literature review confirm that VOCs are one of the most important types of urban air pollution and have both significant environmental and health effects. The most significant source of VOCs includes: traffic emissions and volatile organic compound emissions from vehicle fuels; volatile solvent emissions from various industrial and craft activities; emissions from household and indoor consumer products; volatile organic compound emissions from biomass and/or waste burning; while biogenic VOC emissions provide a natural, seasonal background in warm-season months (Kansal, 2009; Gentner et al., 2017; Pandey & Yadav, 2018; Seinfeld & Pandis, 2016). Meteorology--temperature, solar radiation, wind speed/wind direction, humidity and atmospheric stability/pressure--has a very large influence on VOC dispersion and reactivity as well as the occurrence of VOC-related photochemical reactions resulting in clear diurnal (i.e., rush-hour peak concentrations and post-rush hour minimum concentrations); and seasonal (i.e., winter and post-monsoon VOC accumulation and monsoon VOC dilution) patterns (Seinfeld & Pandis, 2016; Mukherjee & Agrawal, 2020; Dimitriou & Kassomenos, 2015). The international and national literature has identified VOCs as causes of photochemical smog and ozone formation and has documented health effects associated with VOCs with benzene being recognized by many agencies including the World Health Organization (WHO), the United States Environmental Protection Agency (USEPA) and others as a known human carcinogen (ATSDR, 2005; Hester & Harrison, 1995; WHO, 2021; USEPA, 2020). While routine VOC monitoring is relatively common globally compared to PM and criteria gas monitoring in India, it remains sparse for VOCs in comparison, which supports the need for localized VOC assessments to evaluate VOC exposure and support VOC management (Pandey & Yadav, 2018; CPCB, 2022). Recent studies indicate that non-linear VOC - meteorology relationships can be captured using machine learning models such as Random Forest, Gradient/Extreme Gradient Boosting, Support Vector Regression (SVR) and Artificial Neural Networks (ANN) and generally perform better than linear baseline models, especially when multiple meteorological variables are used in conjunction (Xiong et al., 2021; Chen & Zhang, 2020; Peng & Sun, 2022; Ambade et al., 2022). Together these results provide context for the present Barrackpore (SVSPA, Mangal Pandey Uddyan) study: A localized VOC study that incorporates meteorology into VOC modeling using machine learning methods to guide practical VOC-related air quality management.

2.8 Research Gap

The large number of VOC studies in India and globally notwithstanding, there are still several significant gaps that have motivated the current study. First, a great deal of Indian research has focused on the major cities of Delhi, Mumbai and Kolkata, while the numerous semi-urban corridors and campus-adjacent settings (such as Barrackpore and SVSPA) have been studied very little; West Bengal's VOC data collection, especially by continuous monitoring, lags behind other air pollutants such as PM and criteria gases, which limits how well the impacts of VOCs can be assessed locally (WBPCB, 2022; CPCB, 2022; Pandey & Yadav, 2018). Second, while most VOC studies have measured VOC concentrations, they rarely fully describe the VOC-meteorology interaction at the fine temporal scales at which it occurs; local-scale micro-meteorological conditions (wind direction and speed, atmospheric stability, mixing height) and the variability in local land use can alter VOC patterns and behavior over time (diurnal and seasonal), therefore necessitating site-specific analyses and comparisons of VOC concentrations between sites that include diagnostics of wind and pollution rose (Dimitriou & Kassomenos, 2015; Seinfeld & Pandis, 2016; Gentner et al., 2017; Mukherjee & Agrawal, 2020). Third, while machine learning has demonstrated a high degree of success in modeling VOC concentrations in global applications, relatively few Indian studies have applied machine learning techniques to VOC concentrations using local-scale meteorological parameters (Ambade et al., 2022; Chen & Zhang, 2020; Peng & Sun, 2022; Xiong et al., 2021). Therefore, the present study (i) characterizes VOC behavior in the vicinity of SVSPA, Barrackpore, (ii) quantitatively relates VOC behavior to local-scale meteorological parameters at high spatial resolutions, and (iii) compares multiple machine learning models to determine the most suitable model to predict VOC concentrations in a semi-urban setting — thereby filling the identified gaps in terms of location, processes, and methodology.

CHAPTER 3

OBJECTIVES & RESEARCH METHODOLOGY

3.1 General

The chapter describes the general methodology of the study used to determine VOC concentrations and analyze the VOC-meteorology correlation at Swami Vivekananda State Police Academy (SVSPA) area; Barrackpore; Mangal Pandey Uddyan.

A good methodology is essential for a study because it allows the researcher to present a structured approach to the research process and thus helps in ensuring that the entire research process will be conducted systematically; consistently; and be easily followed.

The overall process adopted in this study begins by identifying the study location and selecting a reliable monitoring station where VOC and meteorological data can be collected. The collected data set then undergoes an organisation and cleaning stage to remove all missing values, errors and outliers. The cleaned data set is then prepared for the analysis phase, the focus of which will be to evaluate seasonal and diurnal variations of VOCs to understand how pollutant levels vary over time and with respect to different weather conditions.

Statistical analysis and forecasting using Machine Learning (ML) models based on meteorological input variables, to predict VOC concentration will also be included within the methodology. Multiple models will be constructed, trained and evaluated against each other to determine which model predicts VOC concentrations most accurately. The combination of the above two processes; statistical evaluation and ML-based prediction will assist in providing a greater understanding of VOC behaviour in the study area.

3.2 Major Objective

The primary purpose of the methodology is to investigate how Volatile Organic Compounds (VOCs) behave in relation to the SVSPA Campus, Mangal Pandey Uddyan, Barrackpore and how VOCs are correlated to other atmospheric factors such as meteorology. This goal will be achieved by completing three objectives:

1. Collect VOC data and data associated with relevant atmospheric conditions (meteorological data) at the nearest Continuous Ambient Air Quality Monitoring Station (CAAQMS) located near Barrackpore.
2. Study the seasonal and diurnal behavior of VOC pollutant emissions and temperature, humidity, wind speed, wind direction and atmospheric pressure.
3. Compare a variety of machine learning based models to predict VOC pollutant concentrations using meteorological input variables.

Each of these objectives directs each stage of the research process from collecting VOC data through developing predictive models for VOC concentrations.

3.3 Methodology

This study employed a methodologically defined series of steps to comprehend how VOCs behave in relation to meteorological conditions around the SVSPA campus. In order to gather high-quality air-quality data at the selected monitoring site, first a location was identified as being suitable for a monitoring station; then the study location was identified. Following that, hourly VOC and meteorological data-sets are gathered and thoroughly cleaned to eliminate missing values, anomalies, and inconsistencies.

Then, once the data is ready, it is organized so as to allow a very detailed examination of its variability with respect to time (both hourly and peak hour variations), and seasonally (pre-monsoon, monsoon, winter, post-monsoon variations).

These examinations will provide an understanding of what the normal pollution cycle is and how weather variability affects VOC concentrations.

Finally, the clean data-set is utilized for the development of models where several machine learning algorithms are developed and tested to predict VOCs in the study area.

The algorithms are mainly ranked by their R² scores to identify the best model for VOC forecasting in the study region.

Overall, the complete process of the study from data acquisition to ML-based forecasting is illustrated in the flow-chart presented below.

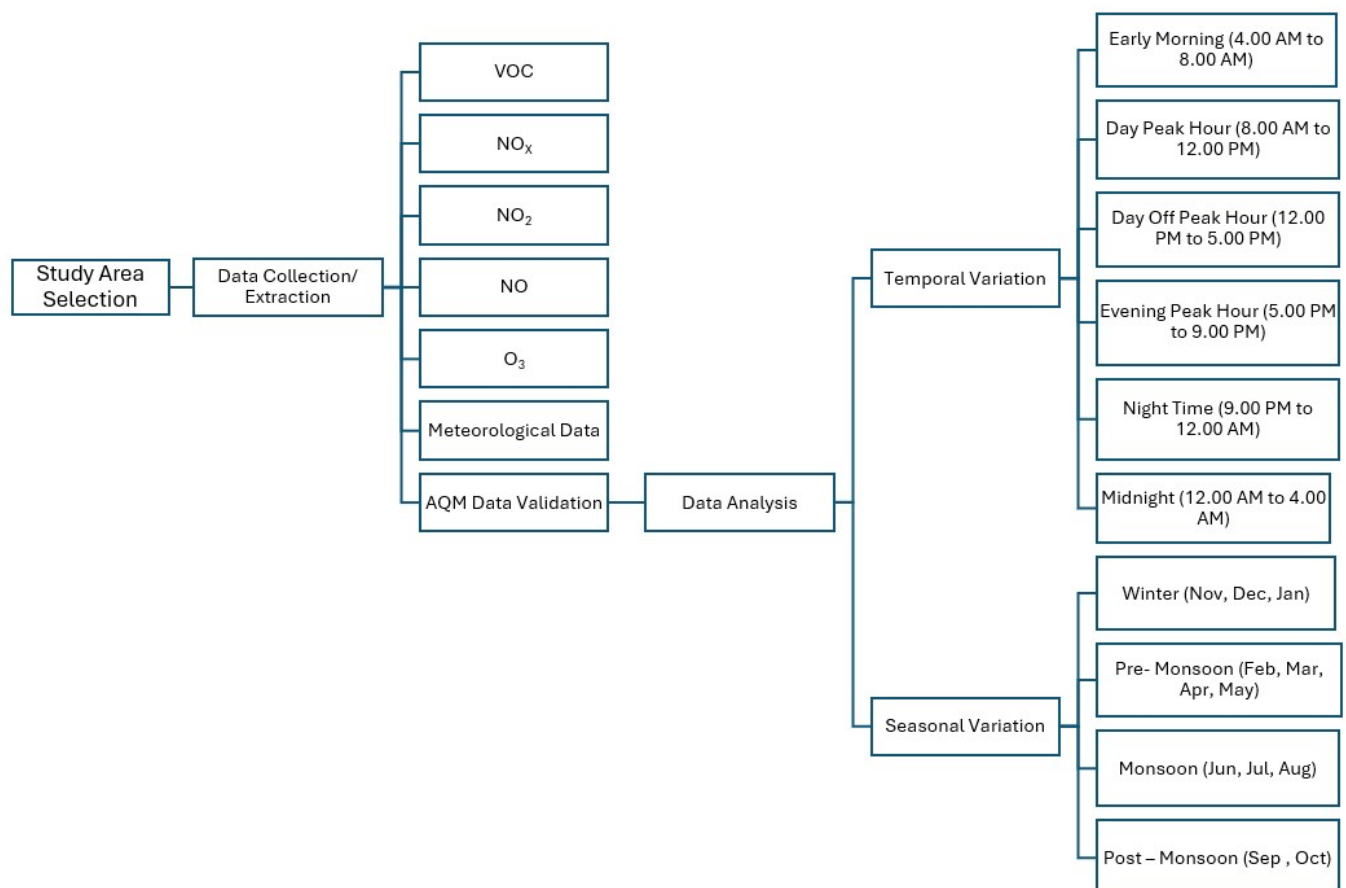


Figure 3.1 — Research Methodology

3.4 Study Area

This study took place around the Swami Vivekananda State Police Academy (SVSPA). Located in the Mangal Pandey Uddyan, Barrackpore; the North 24 Parganas district, West Bengal. It is an area that includes both urban and rural zones. The area is also used for a variety of purposes, including residential areas, training sites, open public parks and has moderate amounts of vehicle traffic each day. Because the surrounding environment receives various amounts of pollution from two wheelers, auto rickshaws, buses and the local market as well as from time to time from building construction, it is an appropriate area to evaluate VOC behaviour.

A CAAQMS (Continuous Ambient Air Quality Monitoring Station) was chosen to provide accurate VOC and other necessary environmental/meteorological data for the area of interest. The closest available CAAQMS to the area of interest was at Barrackpore (BKP). Hourly VOC data as well as hourly meteorological data including temperature, relative humidity, wind speed, wind direction, and barometric pressure were provided by the Barrackpore CAAQMS. Due to the fact that the SVSPA campus is located in the same atmosphere zone as the Barrackpore CAAQMS, the data collected will be representative of the air quality in and around the SVSPA campus.

Geographic Coordinates (CAAQMS Barrackpore)

Latitude: 22.7605581° N

Longitude: 88.3617589° E

From the satellite imagery we see that the CAAQMS is located among trees and near to roads and open spaces so it is a location where both anthropogenic and natural influences are present. The CAAQMS would be a suitable location from which to assess VOC concentrations and how they interact with the local weather.

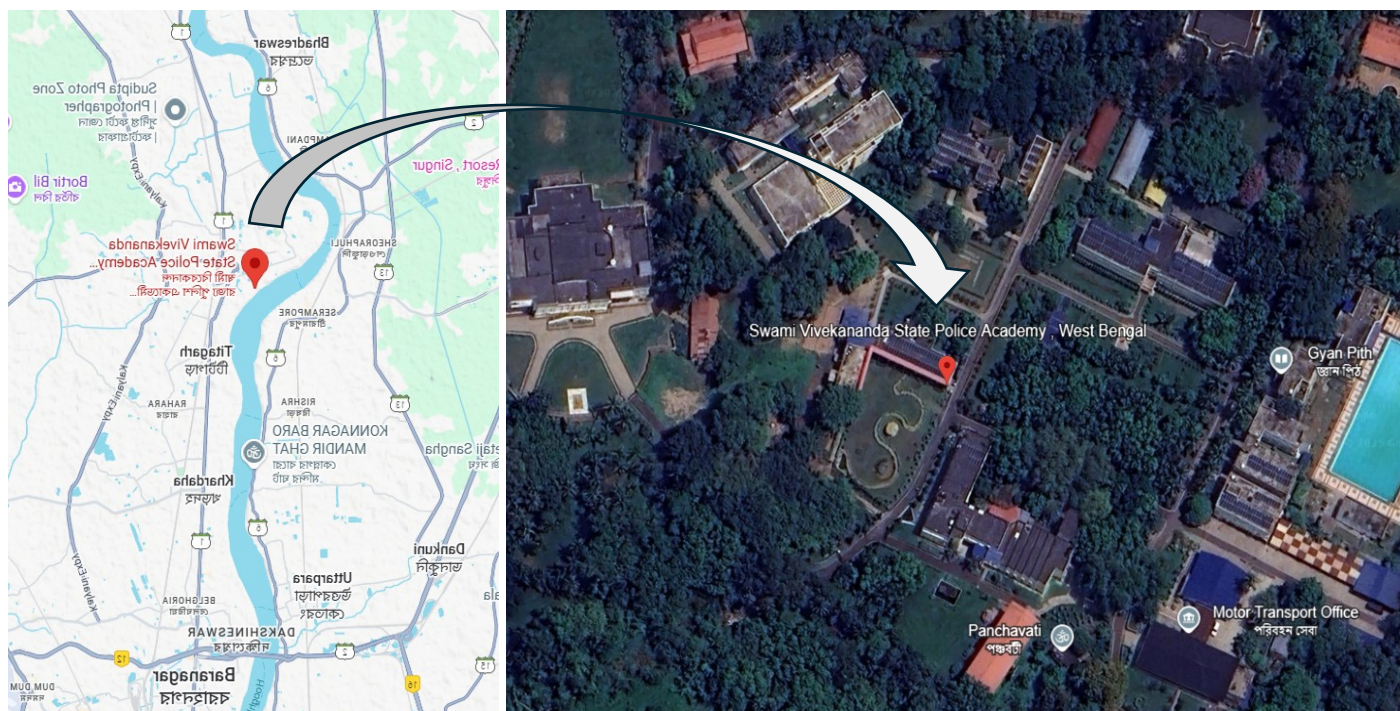


Figure 3.2 — Study Area Map

3.5 Data Collection

Consequently, we retrieved ambient air-quality data used by the nearest continuous monitor in Barrackpore CAAQMS (BKP) that is nearest to Swami Vivekananda State Police Academy (SVSPA) in Mangal Pandey Uddyan. The aim was to get the same level of boundary-layer condition when it comes to exposure within and around campus. The location is semi-urban where the human activity is stable, and land use is mixed, and flows of trainees, staff, residents, commuters, and autos, buses, two-wheelers and occasional construction causes a realistic combination of traffic- and solvent-producing VOC emissions on a biogenic background. Such a representativeness criterion as coherent meteorology, close source mix and short-range transportation, justifies the use of BKP data to estimate the local VOC dynamics at SVSPA (CPCB, 2022; Gentner et al., 2017; Mukherjee and Agrawal, 2020). VOC species (BTEX) were also accessed on the CAAQMS portal and 15-minute observations along with meteorological parameters air temperature (AT), relative humidity (RH), wind speed (WS), wind direction (WD) and barometric pressure (BP) were extracted and native timestamps (IST) and site codes were retained to make things traceable (CPCB, 2022; WBPCB, 2022). The highfrequency series permit us to disaggregate diurnal cycles, seasonal differences and directional behaviour through wind/pollution-rose diagnostics and they serve as inputs to data-driven prediction models (Dimitriou and Kassamenos, 2015; Seinfeld and Pandis, 2016).

3.6 Data Preprocessing

In accordance with the general network advice, screening, harmonisation, and validation preceded analysis and modelling (USEPA, 2020; EEA, 2022; CPCB, 2022). We audited the data and filtered incomplete data and the non-physical data -negative VOCs, impossible WS/WD, duplicate timestamps and blatant telemetry artefacts -flat-lining, datetime rollovers. The normalisation of timestamps and the alignment of the timestamps across pollutants and meteorology were performed to construct a single time-synchronised table. Then we used operational ranges vs. instrument specifications and literature climatology of the eastern Indian area; abrupt and isolated spikes were identified with simple robust rules (e.g., IQR/Hampel thresholds) and omitted in model training and retained authentic peak events (EEA, 2022; Seinfeld and Pandis, 2016). Complete cases were only used in descriptive statistics, and the gaps may be pointed out allowing short gaps to be bridged by conservative interpolation, although we did not gap-fill any ML targets in exploratory plots. Variables were unit-consistent and we came up with categorical encodings (e.g., hour-of-day, season) of diurnal/seasonal analyses. We lastly divided the dataset into analysis subsets (seasonal/diurnal/rose diagnostics) and modelling folds (train/test) to have the ability to consistently assess learning algorithms. The result of this workflow is a clean, time-consistent, analysis-ready dataset which we can analyze to evaluate the VOC-meteorology coupling and train non-linear predictors in a semi-urban environment (Gentner et al., 2017; Mukherjee and Agrawal, 2020; USEPA, 2020).

3.7 AQM Data Validation

Prior to beginning our analysis, it was essential to validate the reliability and suitability of the air-quality data gathered from the Barrackpore CAAQMS station because the data is central to studying VOC behavior and developing our machine learning models. In order to perform an initial validation on the data, we ran a simple validation test to determine its integrity and quality.

Our first task in validating the data was to ensure that the Barrackpore station was functioning throughout the study duration and collecting data every fifteen minutes as planned. We verified whether the time stamps in the data were sequential and whether the sensors were operational continuously with no extended periods of inactivity (gaps) or missing readings. The purpose of this verification was to establish that the data accurately reflected the true environment as opposed to a fabricated break.

Subsequently, we evaluated the pollutant and weather-related data to ensure all parameters fell into acceptable ranges. We tested whether the temperature, relative humidity, wind speed and pressure displayed natural fluctuations as opposed to extreme spikes. For VOC data, we confirmed that there were no negative readings or aberrations that may have indicated sensor malfunction or data degradation.

Finally, we reviewed the structure of the dataset to confirm that each parameter was time-stamped uniformly and that no temporal misalignment occurred during data downloading or extracting. Due to the fact that CAAQMS stations are maintained by authorized environmental agencies, our primary concern in this validation phase was ensuring data consistency as opposed to recalibrating our own data.

In doing this initial validation, we established the dataset's trustworthiness, cleanliness and preparedness to support our diurnal analysis, seasonal comparisons and machine-learning predictive modeling. By performing this validation, we increase the confidence level in the results of our research and provide a high degree of credibility to our findings based upon the data.

RESULTS AND DISCUSSION

4.1 General

In this chapter we present the outcomes of an analysis of VOC (volatile organic compound) and meteorology data collected every fifteen minutes for one year at the Barrackpore CAAQMS (Central Air and Water Quality Monitoring Station) monitoring site close to the SVSPA (Sarojini Naidu Vanita Vishyaalya Postgraduate Institute) campus. Our primary objective in this chapter was to illustrate how VOC levels fluctuate on a range of temporal scales and react to changes in the weather. This was accomplished by showing the general trends in VOC concentration in selected plots and tables that illustrated diurnal and seasonal VOC concentration trends and the relationships between VOC concentrations and meteorological variables (e.g., temperature, relative humidity, wind speed, wind direction).

We also developed and compared the performance of two types of machine learning algorithms – i.e., Support Vector Regression (SVR) and Random Forest (RF) – to determine the potential for these algorithms to predict VOC levels based on meteorological input variables. All the most significant figures are found in the main body of the chapter; the remaining figures and additional analyses can be found in the appendix. The findings in this chapter will help us better understand how pollutants behave in the vicinity of the SVSPA and facilitate our interpretation of the forecasts produced later.

4.2 Trend of VOC

The purpose of this section is to describe general patterns and behaviors in VOC concentration near the SVSPA campus for all time periods using the 15-min CAAQMS VOC data. Trend analysis has been used to identify how VOC concentrations vary with hour of the day (diurnal), month, and with varying weather/meteorological conditions; and these patterns are primarily influenced by traffic movement, localized activity, weather conditions and atmospheric mixing.

Therefore, VOC data were analyzed to examine diurnal behavior/curves and 24-hr variation plots as well as season/monthly plots to determine VOC concentration patterns. Diurnal behavior indicates when VOC concentrations increase/decrease over the course of the day, and are primarily caused by increased morning and evening traffic volumes. 24-hr variation plots indicate when short-term VOC fluctuation events occur, and when sudden spikes due to localized VOC emissions events occur. Seasonal/monthly trends indicate how VOC accumulates/dilutes during winter, monsoon and post-monsoon periods. Therefore, collectively these observations illustrate the general VOC concentration patterns that exist throughout the study area for various environmental conditions.

4.2.1 VOC vs Diurnal Pattern

VOCs vary daily in response to diurnal variations of human activity and short-term changes in meteorology. The graph above illustrates the 15-minutes time series data of VOC diurnal variation. The first rise is observed in the morning as a result of an increase in traffic and the amount of people (students/employees) moving around the SVSPA campus during training and working hours. After the initial rise, VOC decreases during the afternoon due to increased atmospheric instability and increasing wind speeds that allow for better dispersion of pollutants.

The VOC concentration also increases at night due to the second traffic peak of the day as both academy vehicle movements and the public move through the nearby roads. VOC concentrations are generally lower and more stable in late-night and early-morning times representing lower emissions from humans and less active human behaviour.

The Diurnal Variation Graph, in general, provides information about how VOC Levels respond immediately to daily behavior and local atmospheric conditions. Observed Peak/Troughs are a good fit for the common VOC Emission Cycles found in Semi-Urban Areas like the SVSPA Campus Area.

4.2.2 VOC vs Monthly-Hourly and Seasonal-Hourly Variation

In addition to the overall trends of VOC concentrations observed from month-to-month (monthly-hourly) and seasonally (seasonal-hourly), the graphs below show how VOC concentrations vary hour-by-hour throughout the month and the year. The monthly-hourly plot shows distinct peak VOC values for certain months, such as winter and post-monsoon months, where VOC concentrations can be elevated when low temperature and low wind speed conditions allow VOCs to accumulate. Conversely, VOC concentrations are generally lower and more consistent from hour-to-hour in the monsoon months, because the frequent rains and increased atmospheric mixing reduce VOC accumulation.

The seasonal-hourly variation also supports this pattern, with the highest VOC hourly values occurring on morning and evening hours of winter days, and the lowest VOC hourly values occurring on afternoon hours of monsoon days. Collectively these two plots help explain how the long term seasonal characteristics of an area, along with the day-to-day weather conditions, act together to affect VOC concentrations near the SVSPA campus.

4.2.3 Time-Slot Variation of VOC

A time slot error bar plot displays the hour-to-hour variations in VOCs over the course of a day at SVSPA. VOC concentrations tend to be higher than average during both the morning hours (04-08 h) and the late afternoon through early evening hours (17-21 h). VOC emissions from commuting activity and limited vertical dispersion of VOCs appear to be responsible for these increases. Low VOC concentrations during the early morning hours (00-04 h) and late evening hours (21-24 h) show a high degree of variation, as VOC concentrations build-up when there is little wind. The lowest VOC concentrations and lowest variability of VOC concentrations occur during the late afternoon (12-17 h), which appears to be the result of both vertical transport by convection and VOC concentration dilution. Overall, the time slot error bar plot mirrors the cyclic nature of human activity and local weather conditions on an hourly basis.

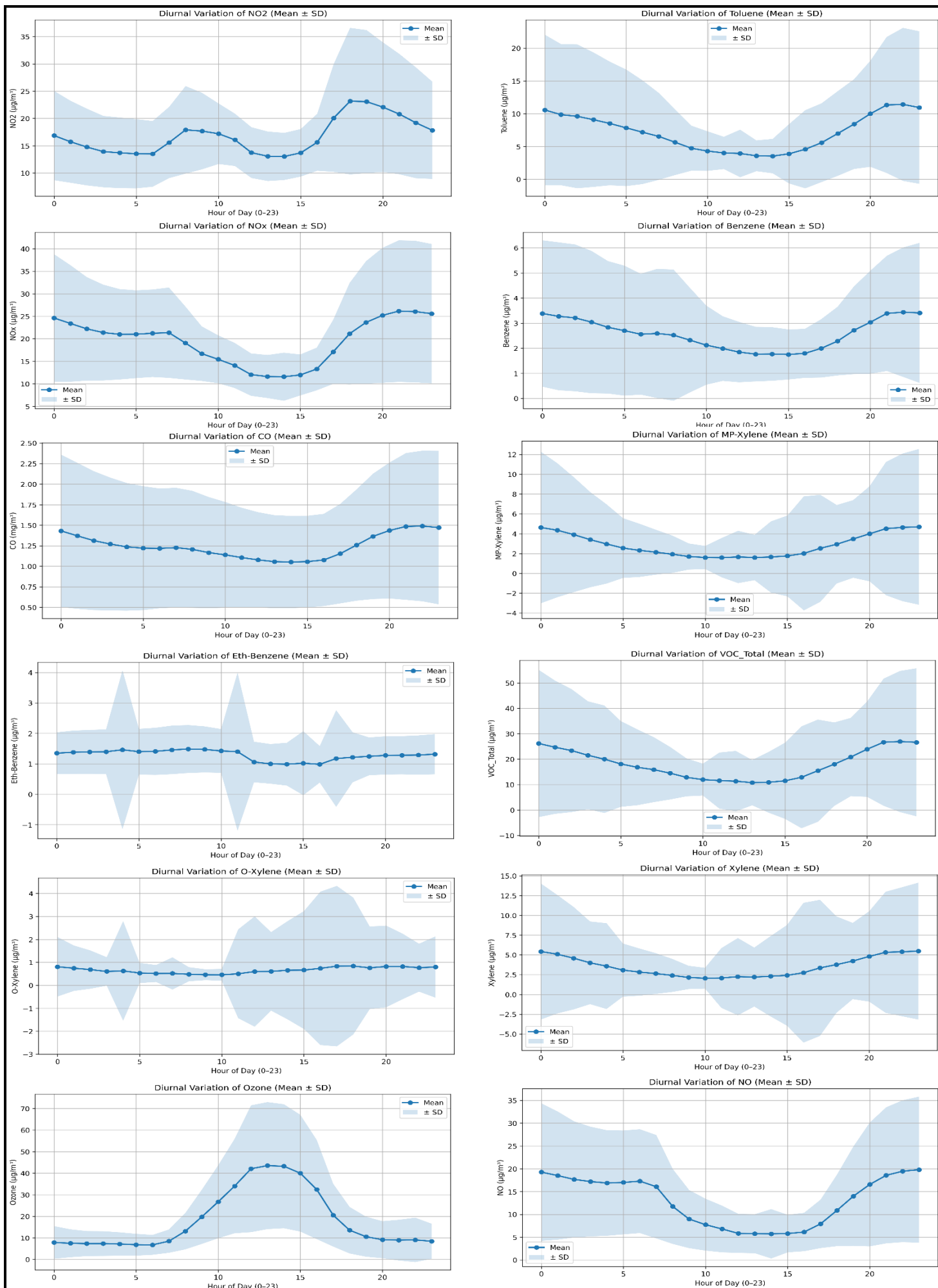


Figure 4.1 — VOC vs Diurnal Plot

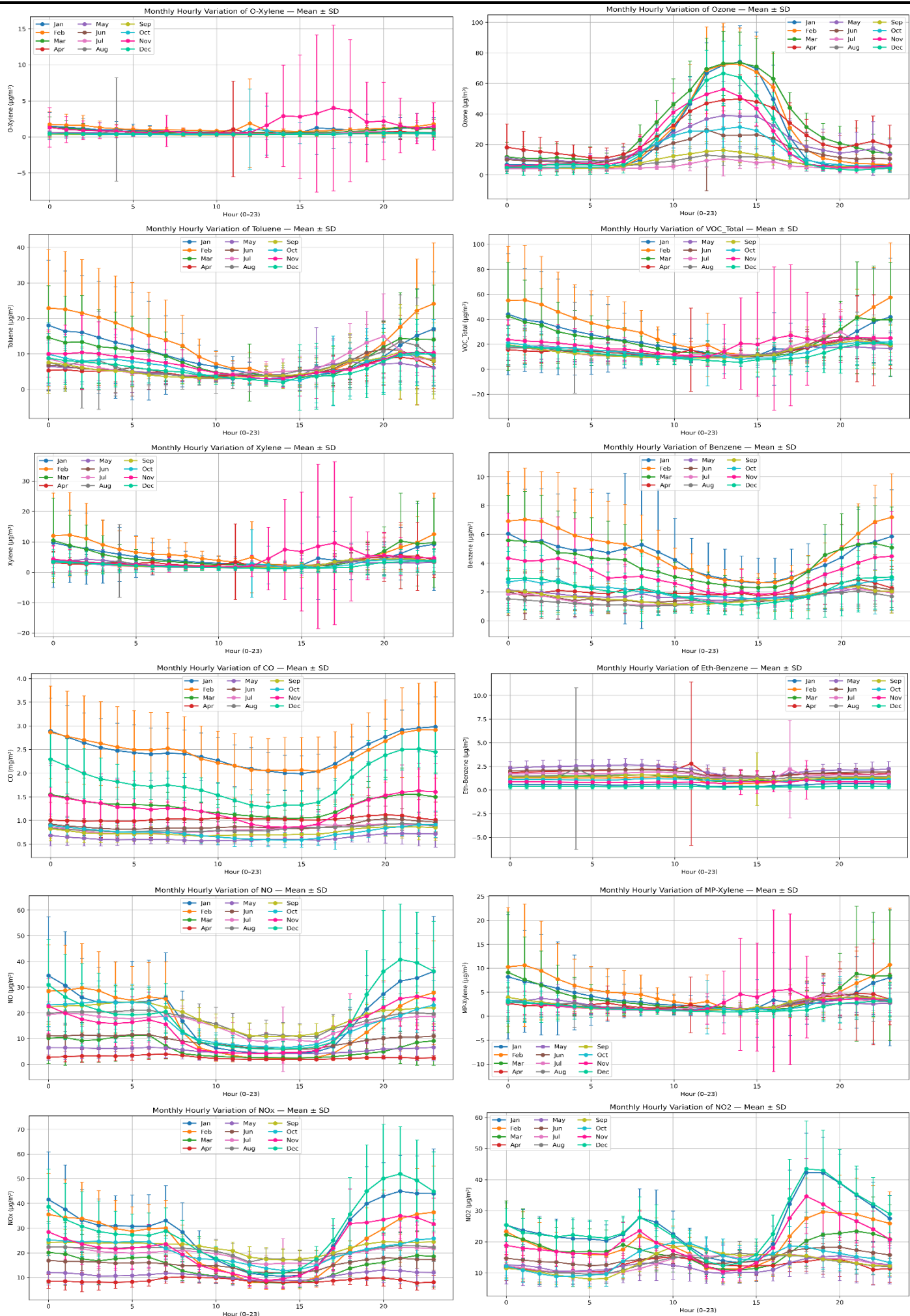


Figure 4.2.1 - VOC vs Monthly Hourly Variation

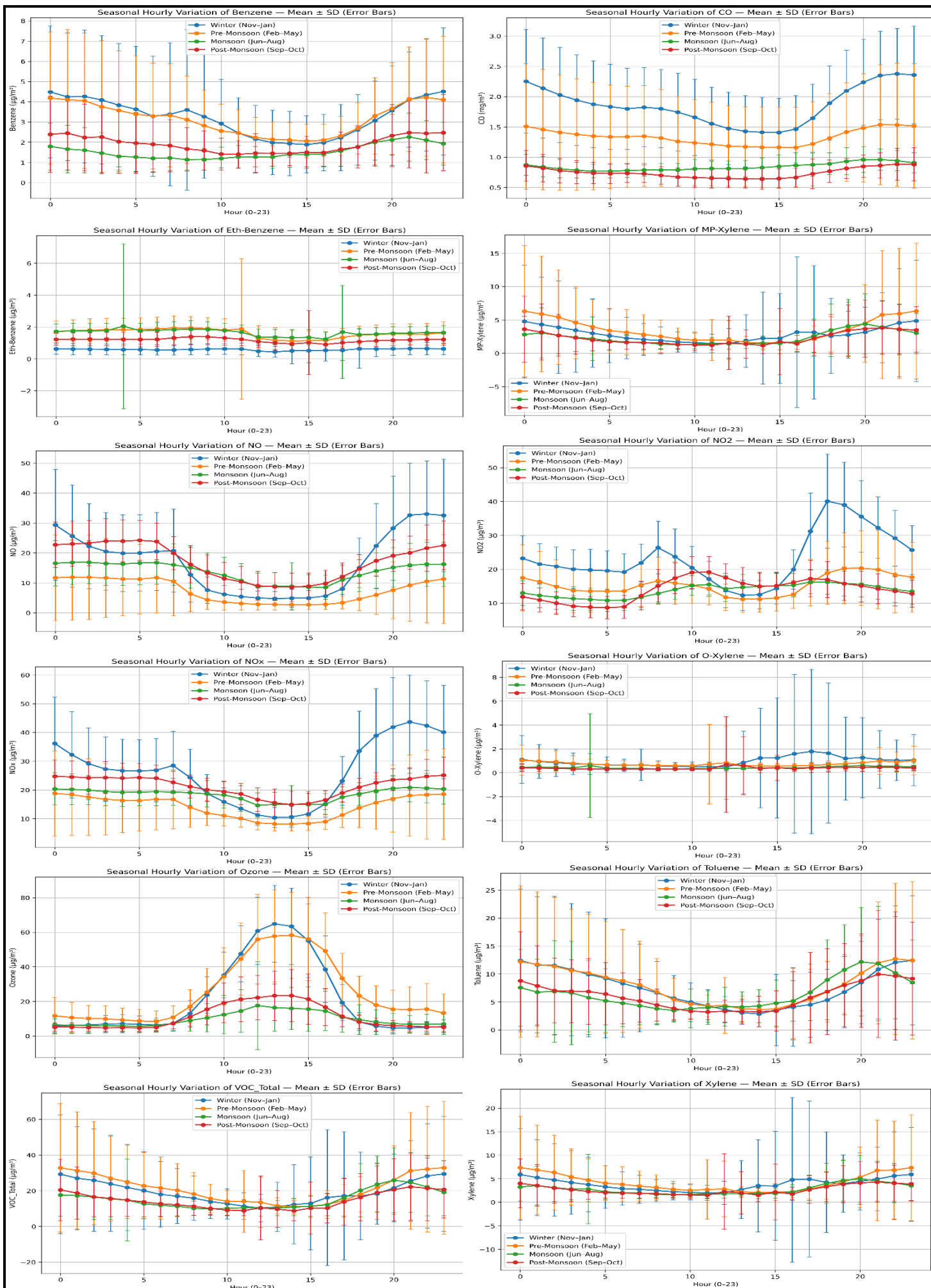


Figure 4.2.2 - VOC vs Seasonal Hourly Variation

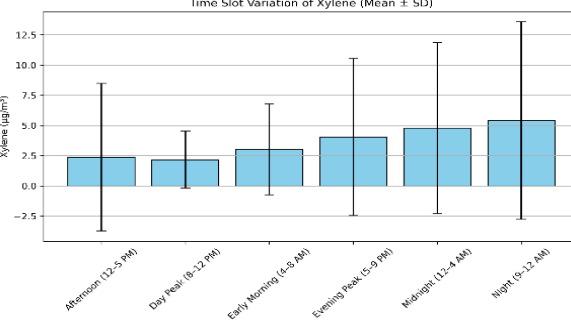
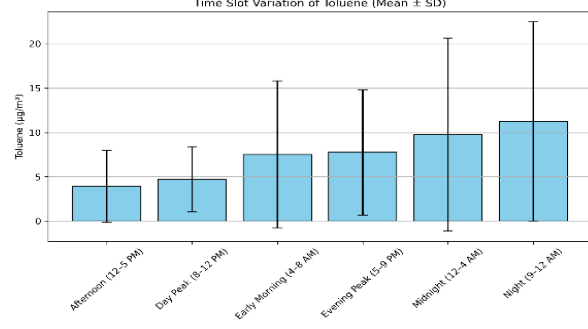
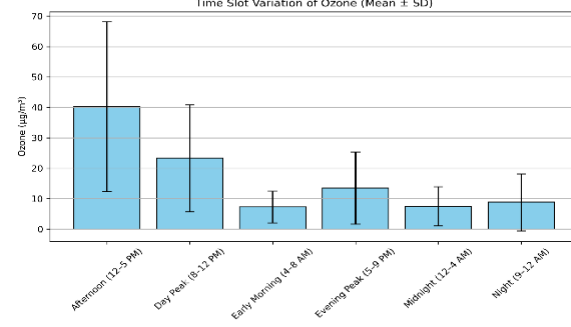
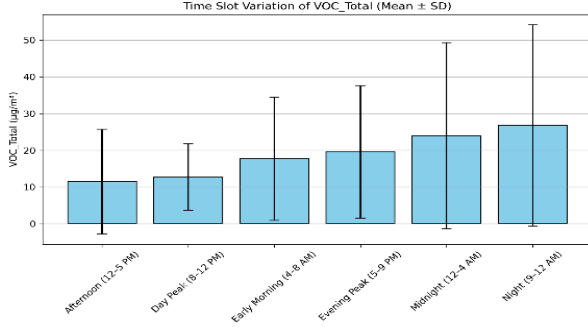
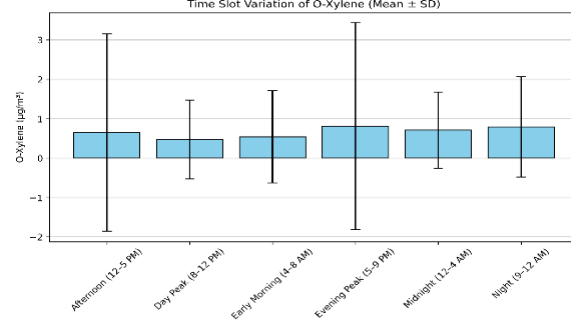
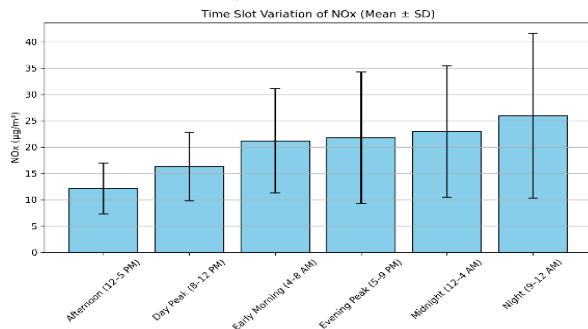
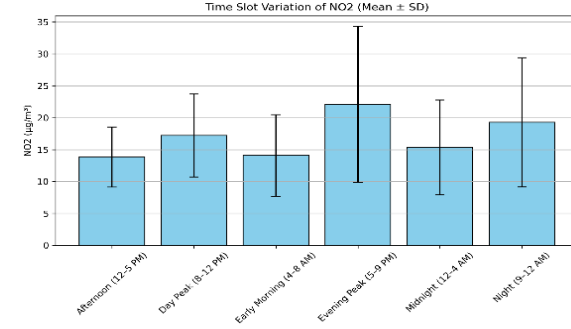
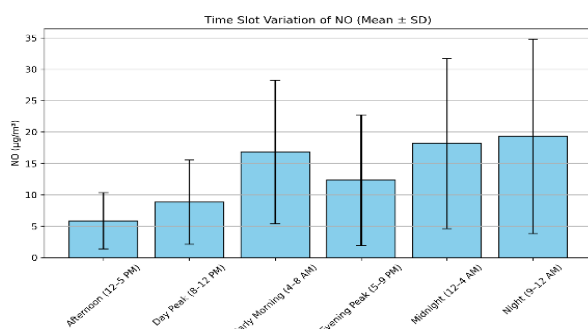
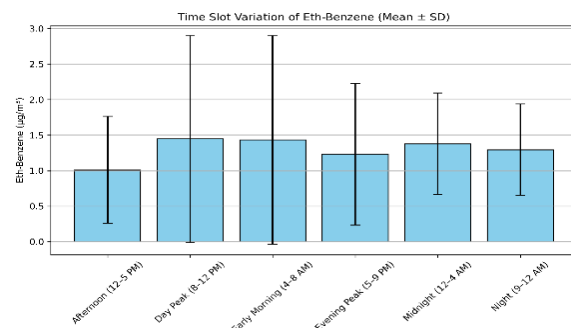
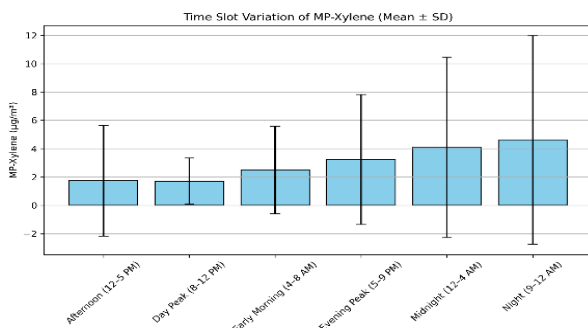
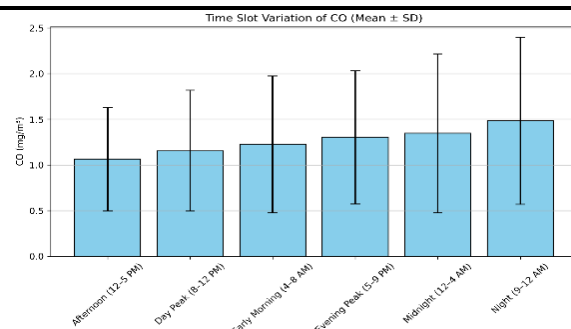
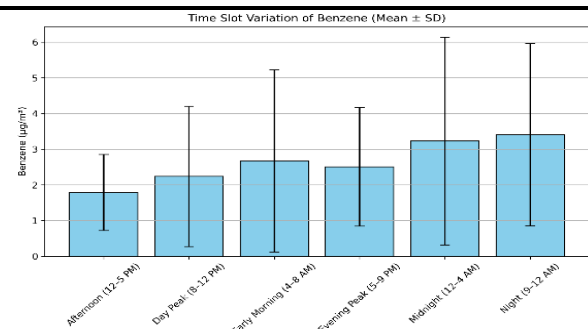


Figure 4.3 — Time Slot Variation of VOC

4.3 VOC vs Meteorological Parameters

Based on the scatter plots of the meteorological variables we investigated — temperature, pressure, humidity, wind speed, and wind direction — it appears there is little or no relationship between those variables and VOC concentrations at the SVSPA campus. This can be inferred because all of the points appear randomly spread out in each plot with low R^2 values, indicating that the relationships are very weak.

The only slight negative trend was observed for VOC vs. temperature; that is, VOC appeared to decrease as the temperature increased. No patterns were identified for VOC vs. humidity and VOC vs. pressure. A mild dilution effect was noted for VOC vs. wind speed, however, it was still too weak to explain the variation in VOC concentrations. Finally, VOC vs. wind direction had no apparent pattern to suggest the source(s) of the VOC emissions were located anywhere specific around the monitoring location. Overall, the results from these plots indicate that the primary cause of VOC concentrations in this area is daily/weekly human activities and nearby emission sources, not from any type of meteorological condition.

4.4 Correlation Analysis and Ozone–NO_x–VOC Relationship

Correlation analysis was conducted to determine the interaction between VOCs and other pollutants and meteorological variables, the methods of correlation analysis included Pearson, Spearman, and Kendall methods. The triangular heatmaps provide a good understanding of the degree to which each of the parameters is connected to the others. The findings of the three types of correlations indicated that the individual VOC species (Benzene, Toluene, Xylene groups) are highly correlated to one another and with the total VOC, as it is expected that these species have similar sources, including traffic and fuel evaporation. There were extremely high positive correlations between VOC_Total and Xylene, O-Xylene, MP-Xylene and Toluene which are the active compounds at the SVSPA campus.

NO_x, NO₂ and CO were also found to be moderately positive correlated to each of the individual VOCs indicating their dependence on combustion and vehicle movement. Conversely, ozone was always negatively correlated with NO_x and VOCs. This is characteristic of the urban and semi-urban areas where the ozone is depleted by the titration of NO, principally during peak traffic periods.

In order to investigate this behaviour more, a 3D surface plot of Ozone versus NO_x versus VOC was plotted. It can be seen on the surface that the ozone concentrations decrease with an increase in the levels of the NO_x and VOC. The maximum values of ozone are only observed in cases where both the NO_x concentrations and VOC concentrations are relatively low. This is in agreement with the common photochemical pattern:

- High NO_x/VOC - Ozone inhibited (titration).
- Reducing NO_x/VOC - Ozone increase (photochemical generated).

On the whole, the plots of combined correlation and 3D surface visualisation demonstrate the conflicting nature of the relation between Ozone and the main pollutants, as well as, they prove that the dynamics of VOC in the area of the study are strongly linked to the emissions associated with combustion.

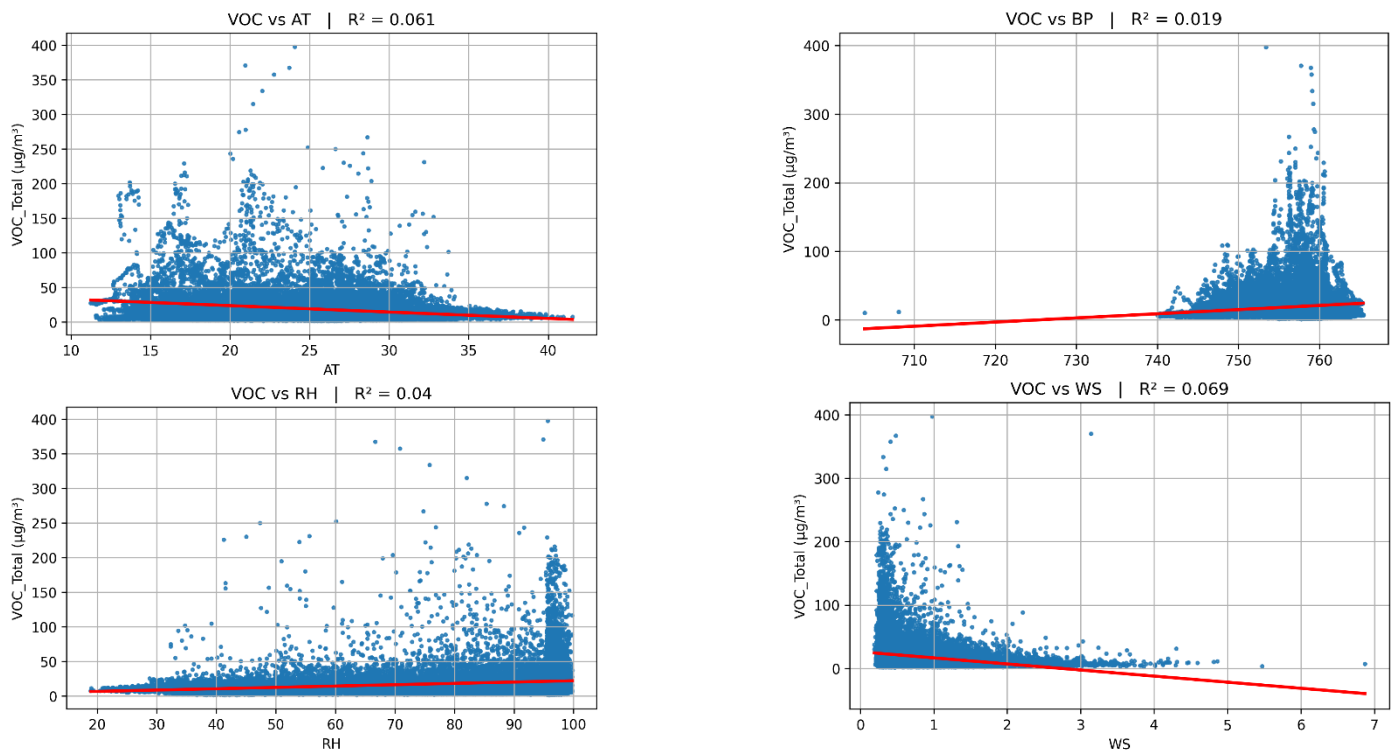


Figure 4.4 — VOC vs Meteorological Parameters

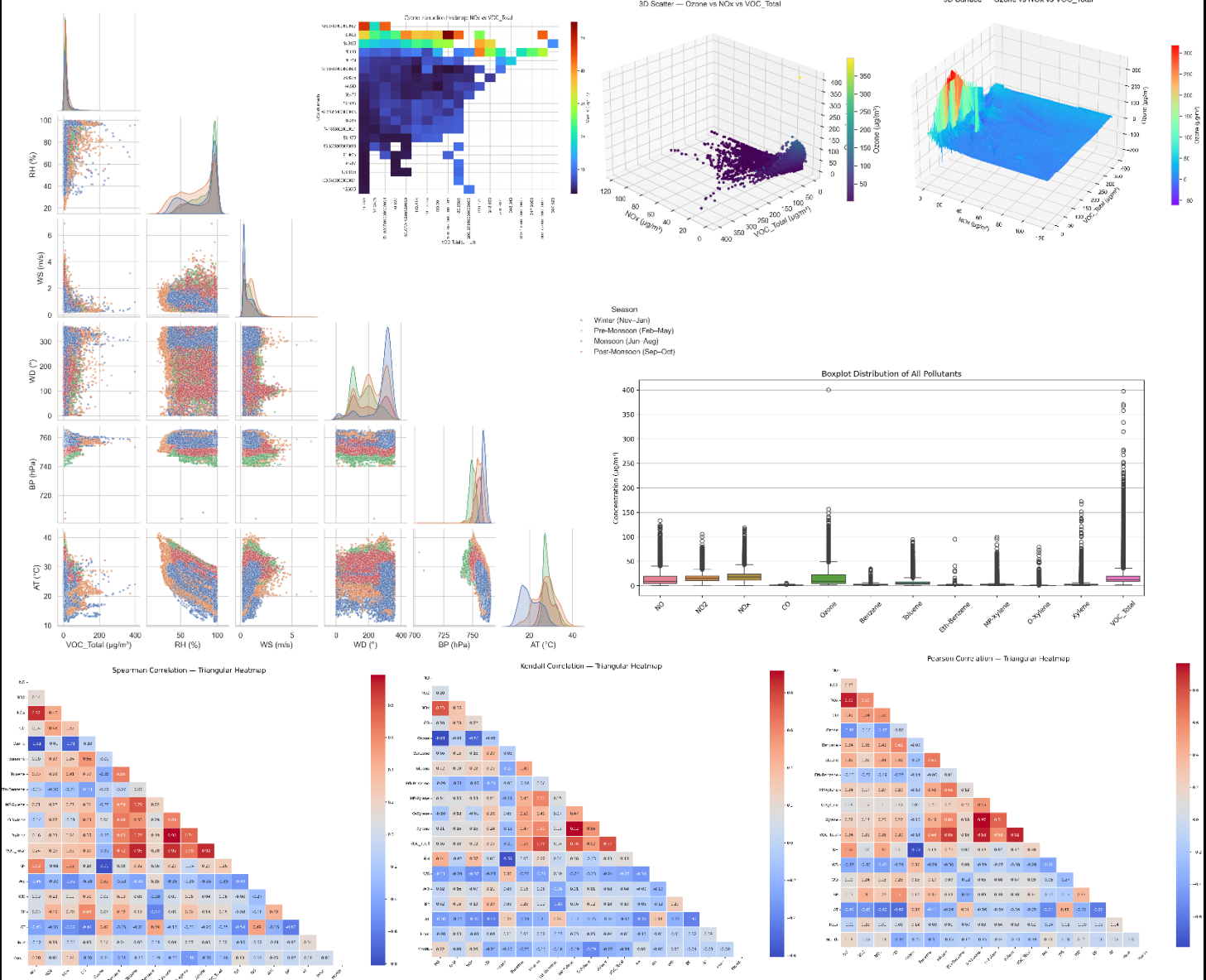


Figure 4.5 — 3D Ozone-NO_x-VOC Surface and Scatter plot, Heatmap, Pair plot, Box plot and Correlation Heatmaps

4.5 Seasonal and Diurnal Wind Roses

Seasonal and Diurnal Wind Rose Plots help us determine how Wind Patterns Affect Pollutant Movement Around the SVSPA Campus. Seasonal Wind Roses illustrate how Wind Direction Changes Throughout the Year. Winter and Post-Monsoon Months Show Calmer, More Consistent Winds Than Monsoon Months Which Have Stronger Winds Coming From the South And Southwest.

Diurnal Wind Roses Illustrate Wind Behaviour Over Different Times Of Day. At Early Morning and Midnight Hours, Winds Are Relatively Slow Which May Cause Pollutants To Accumulate Near The Surface. However, Afternoon And Day-Peak Periods Show Higher Wind Speeds And Greater Dispersion Which Is Associated With Lower VOC Levels Observed During Those Hours.

Overall, The Ten Wind-Rose Plots Provide Clear Visual Representations Of How Wind Speed And Direction Change Based On Time Of Year And Time Of Day To Explain When Pollutants Will Be Most Likely Dispersed Or Remain Concentrated Near The Monitoring Site.

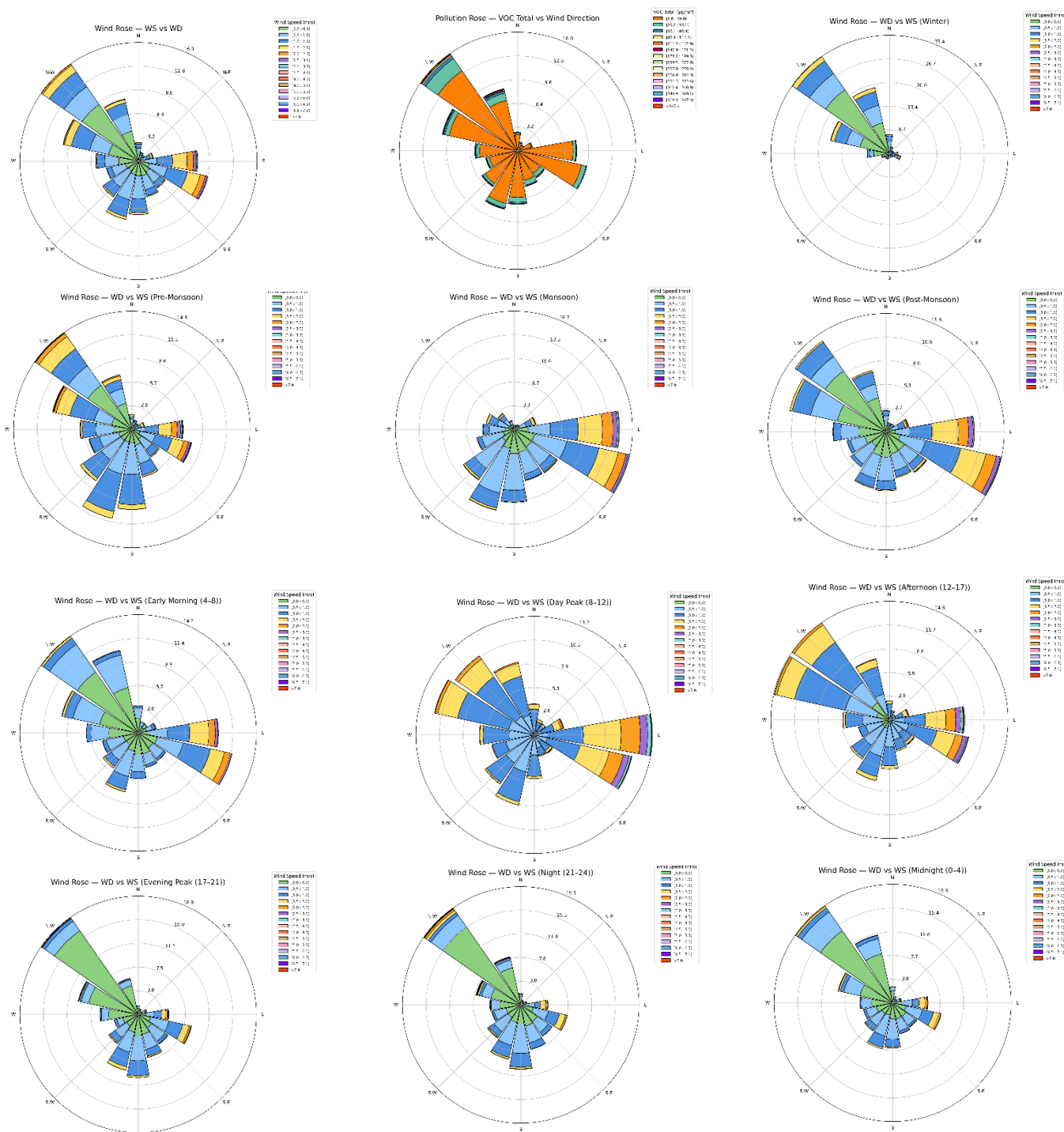


Figure 4.6 — Seasonal and Diurnal Wind-Rose Patterns, Pollution Rose

4.6 Development and Evaluation of Machine Learning Models

This section of the paper formulated and tested a series of machine learning (ML) systems to estimate the **total VOC concentration** using meteorological factors (WS, WD, RH, BP, AT) and co-pollutants. There were seven trained models, including **Linear Regression, Decision Tree, Random Forest, Support Vector Regression (RBF), XGBoost, RNN Variant, and LSTM**.

On a fair comparison all the models were trained on 70% of the dataset and tested on the remaining 30% using the same preprocessing.

To ensure that the chapter does not exceed the figure limit, one combined diagnosis figure was provided: **Figure X Residual vs Predicted VOC All ML Models**.

This number assists us in graphical comparison of the behaviour of each model particularly in the higher ranges of VOC.

4.6.1 Model Performance Overview

All the experiments showed that tree-based ensemble models always performed better than linear and neural networks. Random Forest turned out to be the most effective followed by XGBoost and classical models such as Linear Regression had definite limitations because of the extremely skewed and non-linear characteristics of VOC data sets.

It was evident in the residual-Predicted plots that:

- **Linear Regression and Decision Tree** have big heteroscedastic residuals, which shows that they do not capture non-linearity.
- **SVR (RBF)** has a higher accuracy, but VOC peaks are underestimated.
- The tightest residual spread is created by the Random Forest, and the low and moderate level of VOCs is accurately represented.
- **XGBoost** exhibits the same behaviour and a little more variance.
- High noise and irregular temporal structure of VOC data are a disadvantage to **RNN and LSTM** models.

This behaviour can be attributed to the air-quality literature, in which ensemble models prevail in predicting the concentration of pollutants in the atmosphere because they are able to model complex interactions.

4.6.2 Selection of Best Model

Based on the trends in the residual and quantitative values (R², MAE, RMSE, and Explained Variance):

The best result was the **random forest** which had:

- o Minimal residual dispersion.
- o Minimal systematic bias
- o Constant variation in all ranges predicted.

XGBoost was the second, which was also doing well, but at times it was over-shooting the high values of VOC.

Therefore, the final model to use in **the prediction of the VOC in this study was the Random Forest**.

4.6.3 Discussion of Residual vs Predicted Figure (Final Selected Figure)

The combined figure (Residual vs Predicted all the models) visibly gives the following important insights:

Compact clustering of points around the zero-line of the RF and XGB models suggests a high level of predictive power.

The significant positive residuals at the high VOC levels in LR, DT and SVR models are indicative of the fact that these models do not capture extreme pollution spikes.

The neural network models (RNN, LSTM) demonstrate a poor performance pattern, presumably because of the hourly, discontinuous, and highly fluctuating nature of VOC emissions.

The distribution of the residuals in random forest is the most balanced, which validates its capability to address the noisy and non-linear correlation between VOCs and the meteorological factors. The figure is a full visual diagnosis tool, which brings all the models in one layout, with which it is easy to compare without violating the number of figures authorized.

4.6.4 Final Model Justification

Random Forest has been selected due to:

It has the capability to define **non-linear interactions**.

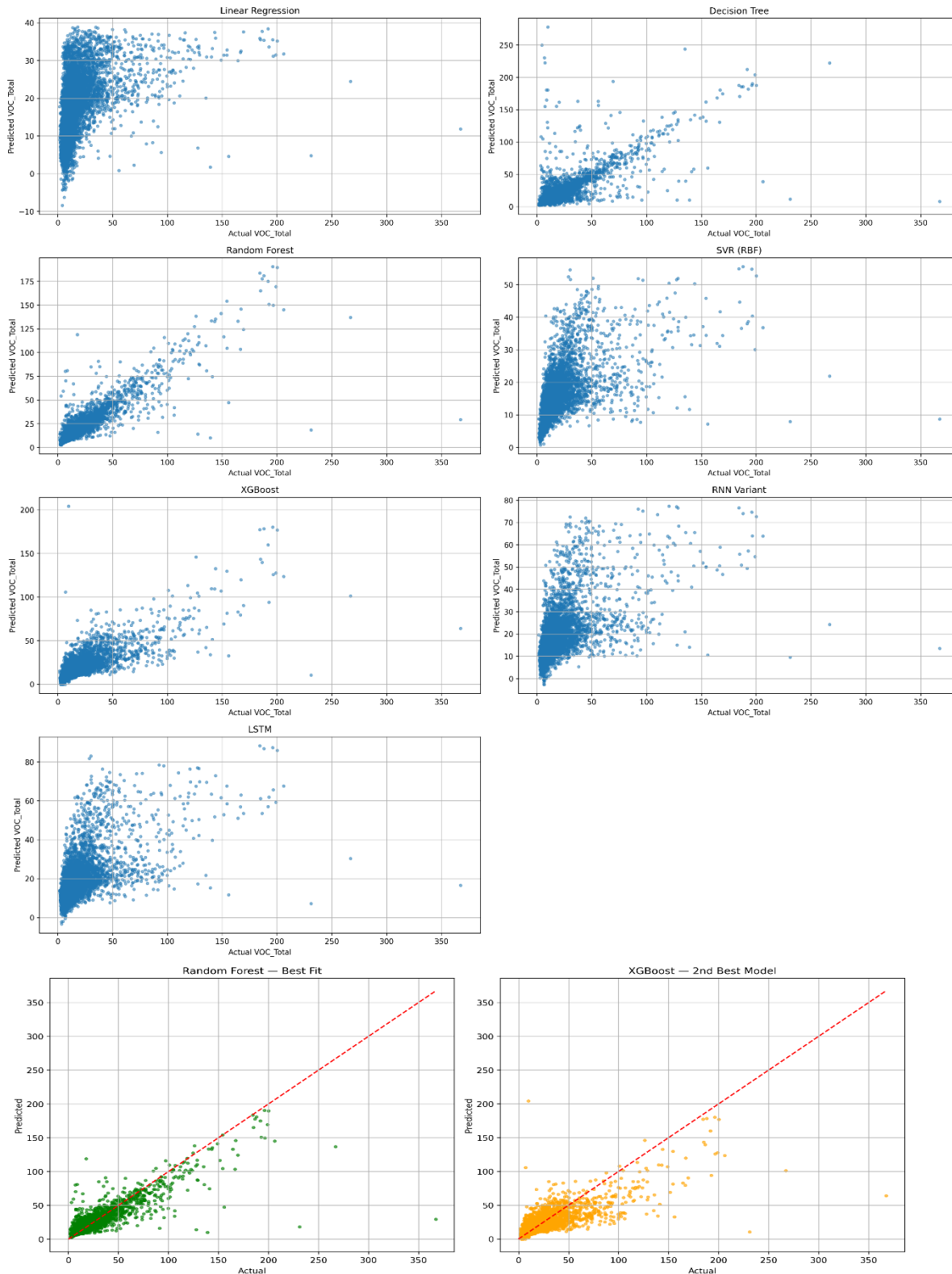
Strong resistance to **outliers and skewed distributions of VOCs**.

Powerful extrapolation of hidden data.

Reduced error measures than any other model.

Optimum residual behaviour of all evaluated models.

Therefore, the final modelling strategy to be adopted in the prediction of VOC in the study site is Random Forest. Robustness against **outliers and skewed VOC distributions**.



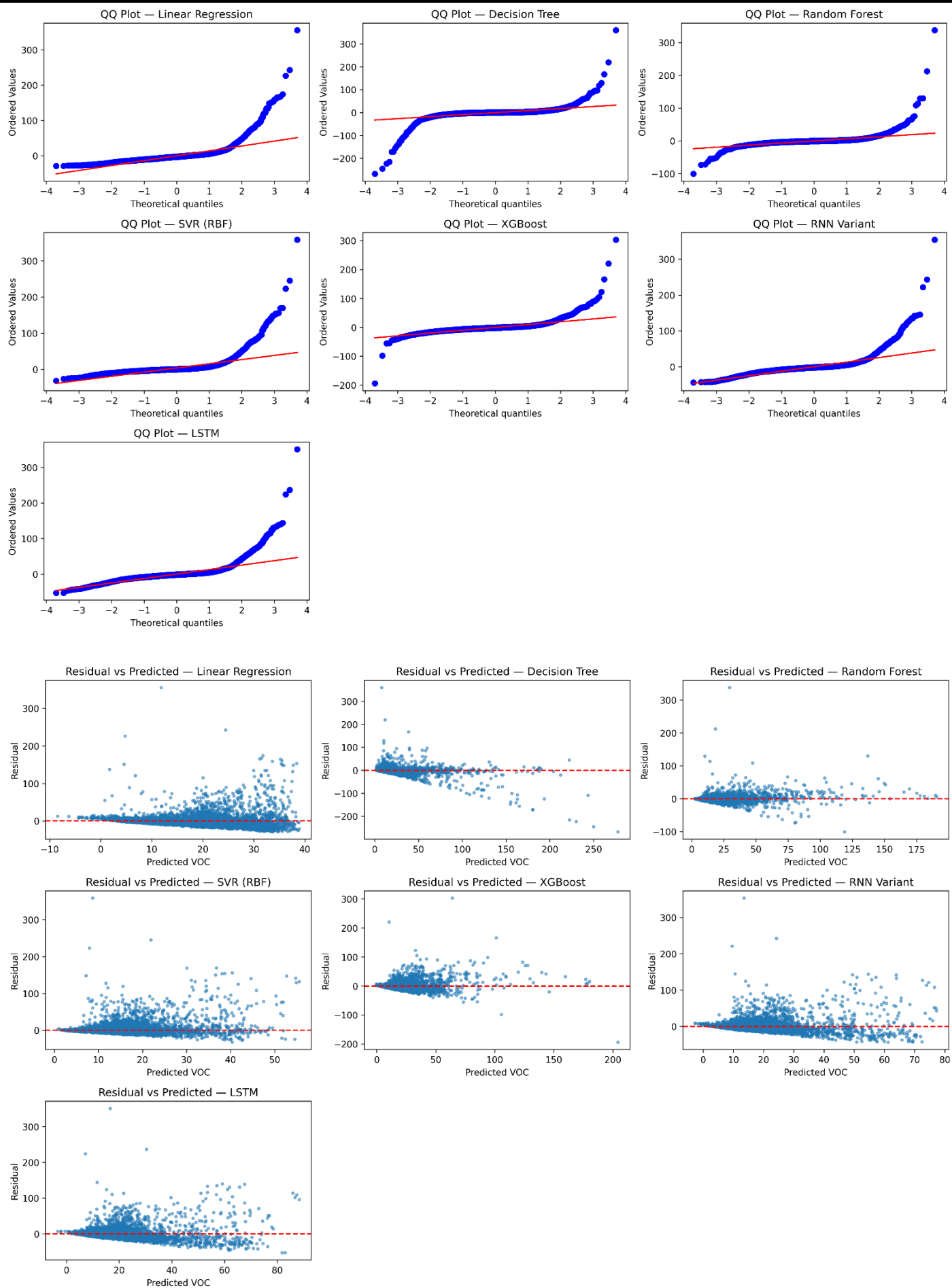


Figure 4.7 — Consolidated Residual vs Predicted VOC Plots for All ML Models (LR, DT, RF, SVR, XGB, RNN, LSTM), illustrating comparative model performance and error distributions.

4.7 Discussion

- **Why We Chose This Project**

Our primary goal when undertaking this project was to know what does in fact affect the level of VOC in a given place such as Barrackpore, particularly, the SVSPA campus, in which human activity, traffic, and mixed land use have a role. Our topic is selected due to the fact that VOCs are disregarded in contrast to PM and NO_x, but it is equally pernicious and is directly associated with the ozone formation and health outcomes. The assignment on this project made us understand the importance of examining the pollutants that individuals may not discuss most of the time.

- **Why We Chose That Area**

We chose the SVSPA Barrackpore neighborhood since it is a highly realistic setting of urban-suburban environment. It is not an industrial center, neither fully residential. Rather it is composed of a balance of mix of the academia activity, surrounding households, local markets, movement of the people by the public transport, and occasional traffic surges. This ensured that the site was perfect in studying the behaviour of VOCs in a realistic situation and not under extreme conditions.

- **Lessons Learned in this Project.**

We found out the extent to which a lot of effort is involved in work with actual environmental data during the analysis. Starting with cleaning of 15-minute datasets and moving on to examine how pollution accumulates or dissipates over a day, each of these steps made us realise how the day washes away the pollution. We also got to learn the effects of meteorological variables on pollutants despite the weak correlation that was discovered in this instance. This is what we learned; atmospheric science is not always so simple - told us sometimes pollutants do something to us that is not readily explained by some simple trends.

- **How This Project Helps Us**

Another key learning experience was the use of machine learning. We understood how models can be used to model the patterns that are not evident at first glance in graphs. The side-by-side comparison of models helped us realize the reasons why certain algorithms are ineffective and why ensemble-based approaches such as the Random Forest are more effective with noisy environmental data.

All in all, this was a technical and academic experience that enabled us to develop. More to the point, it gave us the understanding of the impact of air pollution on actual communities, and how the scientific analysis can help with the future planning and decision-making.

FINDINGS AND CONCLUSIONS

5.1 Key Findings of the Research on VOC Pollution Assessment In Barrackpore Area

The diel patterns of VOCs in Barrackpore have been shown to be strongly influenced by meteorological and land-use factors with consistently higher concentrations found at night and early morning hours. The increases during these hours can be attributed to reduced atmospheric mixing, decreased wind speed, and increased localized emissions from the surrounding area of the SVSPA campus.

In general, VOC concentrations were lowest in the afternoon hours because of increased convective mixing and greater dispersal of pollutants. This was true for multiple months of data collection.

The seasonal behavior of VOCs has shown that VOC concentrations were at their highest in winter with a calm environment, low boundary layer heights and a stable atmospheric condition. Conversely, the lowest VOC concentrations occurred in the monsoon months with rainfall inducing VOC scavenging and an enhancement of dilution.

Correlations between different VOC species (toluene, xylene, o-xylene, mp-xylene) were strong and indicated that there are common emission sources of these compounds including vehicle exhaust, solvents used in industrial processes and evaporative loss from fuels.

Meteorological parameters did not correlate well with VOCs (as evidenced by the low r^2 values) when evaluated on an individual basis. Therefore, it appears that VOC variations in Barrackpore are largely driven by local emission factors and less so by weather factors.

A three-dimensional ozone – NO_x – VOC surface plot indicates that ozone concentrations decrease rapidly as both VOC and NO_x concentrations increase. This relationship is consistent with typical photochemical reactions that occur in urban environments where the reaction between NO_x and VOC suppress ozone production during high traffic periods.

Wind rose plots identified directionally dependent influences on VOC concentrations in Barrackpore with winter and post-monsoon having significantly more stagnant wind conditions and monsoon having significantly stronger south-south-west winds that facilitate dilution.

Machine learning evaluations provided evidence that Random Forest and XGBoost produced better accuracy, more consistent and accurate predictions, and tighter residual distributions relative to linear regression, decision tree, SVR, RNN, and LSTM models.

Random Forest was determined to be the most reliable model overall, providing the best trade-off between prediction accuracy, model robustness and predictability over all VOC concentration ranges.

5.2 Conclusion

By utilizing high resolution (15 minute) time-series air quality and meteorology data in and around the SVSPA Barrackpore Region this research has been able to investigate VOC behavior within the SVSPA Barrackpore Region in detail. It was determined through the analysis of VOC concentration trends over time in the area that VOC concentrations were greatly influenced by emissions from local activities (traffic; institutional; commercial/residential). Trends of VOC concentrations were analyzed on both diurnal and seasonal scales to determine which factors influence VOC concentrations at various times of the day and month.

Correlational analysis of VOC concentration trends was used to identify VOC species with similar sources of emissions and found that none of the meteorological factors evaluated were capable of explaining changes in VOC concentrations over short time frames. An evaluation of the relationship between ozone and NO_x/VOC indicated that there is typical photochemical behavior occurring in urban areas where the formation of ozone is reduced when the levels of NO_x and VOC are elevated.

Through the machine learning model portion of the research it was determined that VOC prediction would require use of non-linear, robust models. A comparison of the results of all models tested revealed that Random Forest yielded the most reliable and consistent VOC predictions and that XGBoost were second best and therefore demonstrate that ensemble learning models can be useful in air-quality forecasting. Ultimately the research met its objective of determining VOC trends; understanding meteorologic influences on VOC; examining VOC and other pollutants' interactions; and evaluating ML models for VOC prediction. The research provides a solid foundation for further studies in air-quality forecasting; environmental management; and urban planning.

5.3 Research Limitations

While this study has provided an insight to VOC behavior at the SVSPA Barrackpore site, it also has many shortcomings;

1. Single monitoring location

The single CAAQMS site analyzed is representative of only that specific location. Therefore, the results could reflect localized conditions and may not be indicative of other locations within Barrackpore, nor even larger areas in Kolkata.

2. Limited number of VOC species

VOC species were limited to those that were measured at the CAAQMS (i.e., benzene, toluene & xylene groups). Other important VOC species (aldehydes, ketones, etc.) were not measured; therefore, the study cannot provide a comprehensive chemical picture of VOCs present in the air.

3. Lack of key meteorological variables

Variables which are known to significantly affect VOC chemistry (such as solar radiation, PBLH, cloud cover and mixing height) were unavailable for inclusion in the study. These limitations reduce the amount of meteorological explanation that can be given.

4. Data gaps, instrument variability

Missing data points, noise in measurements and variable performance of sensors can all potentially reduce the ability of the study to accurately detect trends, the strength of correlations between VOCs and the accuracy of models.

5. Short time frame of study

The short time frame of the study will not allow for the study to capture long term changes (or extreme events) that would potentially have different impacts on VOC behavior (for example, during festivals, or under extreme weather conditions).

6. Machine learning constraints

The machine learning models used in the study were trained using limited amounts of both spatial and temporal data. The absence of emission inventories, and additional pollutant precursors, means that some of the underlying mechanisms controlling VOC behavior may not have been captured.

7. No source apportionment analysis performed

Although the study has identified behavioral patterns in VOCs, it did not perform any quantitative source apportionment analyses to determine which emission sources are responsible for these patterns (for example, traffic emissions, biomass burning, solvent use). These limitations should be taken into consideration when interpreting the results of the study, and when planning for future studies.

5.4 Future Scope

There are numerous opportunities to expand upon this study, both to improve the VOC and total air quality analysis in the study region, based on the results and limitations of this research:

1. Multiple Monitoring Station Expansion:

Including VOC data from multiple CAAQMS stations located in Barrackpore and Kolkata will allow for the creation of VOC spatial distributions and enhanced regional-level VOC conclusions.

2. VOC Species Addition:

Including more VOC species that are chemically diverse, such as formaldehyde, acetaldehyde, isoprene and other hydrocarbon VOCs will provide a greater insight into VOC chemistry and their influence on ozone production.

3. Solar Radiation and Atmospheric Boundary Layer Height:

By including advanced meteorological parameters, i.e., solar radiation, PBLH, CF, PR, and MH, will assist in identifying deeper atmospheric processes and influences on VOC behavior.

4. Creation of a Real Time Predictive Model System:

As Random Forest was the most accurate performing model, a real-time predictive model dashboard or web-based application can be created for providing early warning systems for poor air quality.

5. Evaluating Other Deep Learning Architectures:

Future studies may investigate alternative architectures, e.g., Transformer attention networks, hybrid deep learning models, and stacked ensembles for the potential to increase model accuracy.

6. Further Investigation of Source Apportionment and Chemical Fingerprinting:

Receptor models such as PCA, PMF, and/or UNMIX can be applied to VOC data collected during this study to determine specific VOC emitting sources and assess the relative contributions of these VOC emitting sources to VOC concentrations in the study area.

7. Integrate VOC Concentration Data into Health Risk Exposure Modeling:

Exposure modeling can be integrated with VOC concentration data collected during this study to provide estimates of short- and long-term health risk exposure associated with VOC exposures, particularly for vulnerable populations, e.g., children, the elderly and police personnel at SVSPA.

8. Support for Policy Development and Urban Planning Decision-Making:

This study's findings can be used to support traffic management, emissions controls strategies, green buffers planning, and smart city environmental decisions.

9. Trend Analysis of VOC Concentrations:

Multi-year VOC trend studies will allow tracking of changes in VOC concentrations over time and assessing the effectiveness of implemented pollution mitigation strategies.

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